University of Cape Coast

Information Literacy Skills:
A Course Book

Edited by

Clement Entsua-Mensah

June 2015

© University of Cape Coast

June 2015

ISBN: 978-9988-2-0602-4

All rights reserved. No part of this publication may be reproduced, stored in a retrieval system, transmitted in any form or by any means, electronic, mechanical, photocopying, recording or otherwise, without the prior written permission of the publisher

Published by:

The Library Publications Committee

University of Cape Coast, Cape Coast, Ghana

Printed by:

University of Cape Coast Press

University of Cape Coast, Cape Coast, Ghana

Contributors

Christian Kofi

Juliana Amavi Opare-Adzobu

Paulina Afful-Arthur

Paulina Nana Yaa Kwafoa

Kwame Akonnoh Boohene

Wilfred Kobina Biney

Korklu A. Laryea

Christopher Kwame Filson

Doris Osei-Agyemang

William B. Baiden

Paul Nunekpeku

Foreword

With the proliferation of information sources and its attendant consequence of information overload, it has become necessary that students are equipped with the requisite skills that will enable them to navigate the complex information landscape. In this regard, the University has found it expedient to introduce students to information literacy skills that will not only enable them to become critical thinkers and, in so doing, succeed academically, but also to become lifelong learners.

As a broad concept, information literacy embraces information, ICT and library skills that enable learners to function effectively. Therefore, one's ability to recognize when and what information is needed, appreciate and understand how information is organized, identify the best sources that could be used to locate and access it, then critically evaluate it and use it effectively is what information literacy is all about. It is much more than the acquisition of traditional library skills.

The information literacy course has been designed to meet the Association of College and Research Libraries (ACRL) information literacy competency standards which are to:

- determine the extent of information needed;
- access information effectively and efficiently;
- evaluate information sources critically;
- use information effectively for specific purpose; and
- useinformation ethically and legally.

This is a one credit course which is offered for one semester and focuses on using the library, and developing the necessary research skills of the students. Essentially, the course is geared towards equipping the students with the skills to effectively access information from both print and electronic sources and critically evaluate and use it ethically and legally for decision making and problem-solving activities.

Clement Entsua-Mensah

Editor

CONTENTS

Chapter One: An overview of information literacy Skills *by Christian Kofi* 1

Chapter Two: Libraries *by Juliana Amavi Opare-Adzobu, Paulina Afful-Arthur, Korklu A. Laryea and Christopher K. Filson* 15

Chapter Three: Information resources and their uses *By Korklu A. Laryea, Christopher K. Filson, Paulina Afful-Arthur and Juliana Amavi Opare-Adzobu* 35

Chapter Four: Organisation of knowledge *by Christian Kofi, Wilfred Kobina Biney, William B. Baiden and Doris Osei-Agyemang* 61

Chapter Five: Ethical issues: Copyright, citation and plagiarism *by Christian Kofi, Wilfred Kobina Biney, William B. Baiden and Doris Osei-Agyemang* 80

Chapter Six: Using the Internet to access information *by Kwame Akonnoh Boohene, Paulina Nana Yaa Kwafoa, Wilfred Kobina Biney and Paul Nunekpeku* 101

Chapter Seven: Searching tools on the Internet by *Paulina Nana Yaa Kwafoa, Kwame Akonnoh Boohene, Wilfred Kobina Biney and Paul Nunekpeku* 120

Author Index 155

Subject Index 157

CHAPTER ONE

AN OVERVIEW OF INFORMATION LITERACY SKILLS

By
Christian Kofi

"Ultimately, information literate people are those who have learned how to learn. They know how to learn because they know how knowledge is organized, how to find information and how to use information in such a way that others can learn from them. They are people prepared for lifelong learning, because they can always find the information needed for any decision or task at hand." (ALA, 1989).

1.0 Introduction

The current emphasis on information literacy as an educational concept is to create literate communities and the fundamental responsibility rests with the librarian to provide access to and delivery of timely information. It is mainly for this reason that library user initiatives such as user education have been developed to support information seeking behaviour. The overall impact of these user literacy education initiatives are geared to critical thinking and problem solving thereby leading to independent learning for life.

Indeed all of us are confronted with making a number of decisions on daily basis. How successful we are able to do this depends on a

combination of skills known as "information literacy." Information literacy means more than being able to read or use a computer. It means knowing how to find, evaluate, and use the best, most current information available to us. Librarians were among the first to identify the importance of information literacy in an information society. In 1989, the American Library Association's Presidential Committee on Information Literacy wrote: "How our country deals with the realities of the Information Age will have enormous impact on our democratic way of life and on our nation's ability to compete internationally." That was before the Internet became an information super highway running through businesses, libraries, schools and, increasingly, homes. The new millennium has brought an urgent need to help schools, campuses and communities understand the changes that technology has brought, and how to reap the benefits and minimize the risks.

In a world that is information rich, librarians are information smart. They are the ultimate search engine. They can help you find the best information, whether it is a website, book, video or pamphlet. Teaching others how to critically evaluate information is a unique skill that librarians bring to a society suffering from information overload. Librarians bridge the information gap by ensuring that people have the resources and skills they need to live, learn, work and govern in democratic dispensations. Librarians provide more than facts. They provide the expertise and services that add meaning to those facts.

Library advocates know libraries are part of the solution, both as resource centres, and as centres for public education. In public, school, academic and special libraries, librarians help to ensure a society where everyone is information literate. The goals of acquiring information literacy skills are listed below:

- Communities enjoy a high quality of life because leaders make good decisions based on sound information.
- Workers have the skills and competencies they need to work effectively in a complex information environment.
- Students graduate with the skills and competencies they need to succeed in the world of work.
- People of all ages and backgrounds have the resources and skills they need to fully participate in democracies.
- Libraries of all types receive increased support as cultural and dynamic centres for information literacy and lifelong learning.
- Librarians are recognised as information experts and key players in the education process who teach others how to navigate a rich and complex, evolving information/technology environment.

1.1 What is Information Literacy?

Over the past three decades, the concept of information literacy has been variously discussed. According to Thomas (2004), Zurkowski in 1974 defined information literacy as "the ability to use techniques and skills from a wide range of information tools as well as primary sources in moulding information solutions to problems". Ten years later, the National Commission on Excellence in Education explained literacy as "the skills required

for new careers and citizenship" and "life-long learning" (Baumbach, 1986). In the mid-1990s, the American Library Association's Presidential Committee on Information Literacy described a literate person as one who can "recognise when information is needed, has the ability to locate, evaluate, and use it effectively and has learned how to learn".

Although most of the above definitions are globally acceptable and therefore applicable, Loertscher (1996) defines literacy in terms of the information needs of children noting that they are avid readers, critical thinkers, creative thinkers, interested learners, organized investigators, effective communicators, responsible information users, and skilled users of technological tools. Later, Montgomery (1997) expanded on the theme, acknowledging that information literacy has become an umbrella concept including electronic searching and information retrieval skills, library skills, media skills, research skills, reference skills, learning skills as well as study skills.

According to Thomas (2004), information literacy may be understood to be as the body of knowledge, skills, competencies and understanding required by an individual to find information effectively and use it appropriately to meet the need that prompted its acquisition.

In sum, information literacy therefore, is the knowledge of when and why you need information, where to find it, and how to evaluate, use and communicate it in an ethical manner.

The Association of College & Research Libraries (ACRL) defines information literacy as "the set of skills needed to find, retrieve, analyze, and use information." ACRL has defined five standards for Information Literacy:

Standard 1: To Know

The information literate student determines the nature and extent of the information needed.

The student must be able to identify these questions:

- What is it you want to know?
- What kind of information do you need?
- How much information do you need?

Standard 2: To Access

The information literate student accesses needed information effectively and efficiently.

The student must be able to identify these questions:

- What is the best way to gather this information?
- Am I using the best terms for this search?
- Which search system or other resource will get me this information?

Standard 3: To Evaluate

The information literate student evaluates information and its sources critically and incorporates selected information into his or her knowledge base and value system.

The student must be able to identify these questions:

- Is this a credible source of information?
- Is there another interpretation or point of view?
- How does this new information change what I know?

Standard 4: To Use

The information literate student, individually or as a member of a group, uses information effectively to accomplish a specific purpose.

The student must be able to identify these questions:

- What is the best method for presenting this information?
- Will this image convey the message I want?
- Are these quotes supportive of my ideas?

Standard 5: Ethical/Legal

The information literate student understands many of the economic, legal, and social issues surrounding the use of information and accesses and uses information ethically and legally.

The student must be able to identify these questions:

- Can I make a copy of this material?
- What are the issues surrounding censorship?
- Are there policies about information gathering, use or reproduction and dissemination?

In order to retrieve the information you need, it is really important to know how the information is organised. Organisation provides access to information or a collection. Libraries useorganisation schemes to 'classify' information according to the academic discipline or area of knowledge (class) the information fits into. The Library of Congress (LC) has developed a system that is used by most academic libraries. Being familiar with the LC system will help you find material in the catalogues of an academic library.

1.2 The Importance of Information Literacy

Information literacy is important owing to the amount of information that is available in contemporary society. Simply being exposed to a great deal of information will not make people informed citizens; they need to learn how to use this information effectively, (ACRL, 2000).

Information literacy is very important because we are inundated with a lot of information in a variety of formats. The same credibility can however, not be attached to all these information. While some may be authoritative, current and reliable, others may be biased, out-of-date, false and misleading. With the advent of information and communication technologies, the amount of information that is being generated is assuming alarming proportions. Therefore the types of technologies required to access, manipulate and create information will definitely increase.

The definition of an information literate person extends beyond school and application to academic problems such as writing a research paper, and reaches right into the workplace. Information literacy is also important to an effective and enlightened citizenry, and has implications that can impact the lives of many people around the globe. The ability to use information technologies effectively to find and manage information, and the ability to critically evaluate and ethically apply that information to solve a problem are some of the hallmarks of an information literate individual. Other characteristics of an information literate individual include the spirit of inquiry and perseverance to find out what is necessary to get a job done.

We live in the Information Age, and "information" is increasing at a rapid pace. We have the Internet, television, radio, and other information resources available to us every day. However, simply because so much information is easily and quickly available does not mean that all of it is worthwhile. With the availability of the Internet, finding high-quality information is now harder than ever before and finding the relevant and quality information is not always quick. Moreover, good quality information does not always come cheaply. To make matters worse, just because you know how to use a particular information technology today does not mean that there is not another one right behind it that you will have to learn how to use tomorrow. At a point, seemingly exotic technologies like "word processing" and "electronic mail" which are now commonplace, were amazing and revolutionary.

Today's employers are looking for people who understand and can adapt to the characteristics of the Information Age. If a student has "learned how to learn," upon graduation, he is a much more attractive job candidate. An information literate individual with strong analytical, critical thinking and problem-solving skills can be considered to be a more adaptable, capable and valuable employee. Information literacy equips us with the necessary skills to recognise when we need information, where to locate it, and how to use it effectively and efficiently. Consequently, it will help in decision-making and productivity which is beneficial to the society.

In the twenty first century, life-long learning has become one of the main themes in higher education. Therefore, the students need to be educated with regard to the abilities and skills of how to learn, or learning to learn, by developing the aspects of reasoning and critical thinking. Information literacy skills will help students to achieve this target in a broader sense, in student-centered learning. Traditionally, the assumption was that the students will gain information literacy skills automatically by themselves, but it is certainly not the case. In fact, information literacy skills need to be inculcated in the students, by the teachers and librarians.

Learning mainly focuses on achieving knowledge, skills and attitudes, associated with particular subject areas. Notwithstanding the subject area, every student should be able to access, use and communicate information in an innovative

manner. An Information Literacy curriculum plays a major role in order to cultivate these skills in university students.

1.3 Uses of Information Literacy Skills

Information literacy skills are used for academic purposes in such areas as scholarly publications and group presentations. The skills identified by scholars and educators as dimensions of critical thinking include the ability to set goals, adjust strategies, carry out tasks, distinguish fact from opinion, establish authority of sources, assess accuracy and relevance, detect bias and underlying assumptions (Thomas, 2004). Undoubtedly, these are the skills that support information literacy skills instruction. Information literacy skills are needed on the job so that one is able to find, evaluate, use and share information. Consumers also need to be information literate so that they are able to make decisions on the choice of what to buy. Information literacy skills are also required in a modern democratic dispensation if one is to be functional in it.

1.4 Evaluating Information Sources/Results of Searches

When you are gathering information, whether it is from books, journals, the Internet, etc., you will need to know whether it is good information. There are many questions you should ask yourself as you decide whether a particular source is good for your research. It is also important to note that the means by which you search for information does not produce misleading or incomplete results. These concern the authority, relevance, coverage, accuracy, currency, quality and context of the material.

Authority and Credibility:

- Who is the publisher?
- Who is (are) the author(s)?
- What makes the author an authority on this subject?
- Does the author cite his/her experience/credentials?
- Does the book or article contain footnotes?
- If so, does material taken from other sources appear to be fully credited?

Scope and Relevance:

- Who is the intended audience? (general, specialized, scholars, etc.)
- What is the level of the material? (basic, advanced, for geniuses only, etc.)
- Is the material appropriate and useful for your purposes?

Coverage:

- What time period is covered?
- What geographical area is covered?
- How comprehensive and specific is the information?

Bias and Accuracy:

- How is the information presented? (fact, opinion, propaganda, etc.)
- If presented as fact, is it accurate?
- Is there a bias? (cultural, political, religious, etc.)
- If so, is the bias clearly stated?

Currency and Timeliness:

- How recent is the book or article?
- How important is this to you and your topic?

Quality:

- What was the quality of the research methodology?
- Did it use accepted methodologies for its field, as far as you know?
- What is the validity
- Is the article clearly written and well-organized?

Context:

- How does the book or article fit into its discipline discourse?
- Is new research being reported or is the author reviewing previously conducted research?
- Does the author agree or disagree with the majority of other scholars in the discipline?
- Is this the first article of its kind within its discipline?
- When you start research on a topic you generally begin with some broad ideas and then refine and focus the search as you begin to gather information. Keep a record of what you find and where you found it because not only will you need it for your project's bibliography, but you may need to re-visit some of the sources as you focus your search.
- There are several steps to this process: identify the topic; gather information; focus the research. You'll probably have to take the steps over and over a few times. As you proceed,

you'll formulate a focused research question. That's the question you'll be answering in your paper.

- Another way to approach a topic is to create a 'concept map' - a brainstorming technique that helps organize a research problem. From there a search strategy can be developed to assist in finding information.

1.5 Ethical and Responsible Use of Information

It is important to note that the information so acquired should be used in a responsible, culturally sensitive and ethical manner. This means that anytime you use somebody's work you have to acknowledge the author. Here, the issues include intellectual property, plagiarism, copyright, fair use, freedom of information, data protection among others. Students have to be prepared to be accountable users and producers of information. Therefore, assignments and project works they undertake must clearly demonstrate their own thinking and not just a conglomeration of "cut and paste" phrases conveniently borrowed and meshed together from numerous sources and of doubtful quality and content.

References

ACRL Information Literacy Web Site Step-by-Step Standards Retrieved from http://www.ala.org/ala/mgrps/divs/acrl/issues/infolit/standards/steps/index.cfm on 21st January, 2014.

American Library Association (1989). Presidential Committee on Information Literacy Final Report. Chicago: American Library Association.

ACRL (Association of College and Research Libraries) (2000). Information Literacy, Retrieved from http://www.ala.org/ala/acrl/ on 21st January, 2014.

Baumbach, D. T. (1986). *Information skills for the information age: The state of our States*, In Aaron, S. L. & Scales, P. R. (Eds.), School Library Media Annual (Vol. 4, pp. 278-285). Littleton, CO: Libraries Unlimited Inc.

Chartered Institute of Library and Information Professionals (CILIP). What is Information Literacy? Retrieved from www.cilip.org.uk/cilip/advocacy on 21stJanuary, 2014).

Loertscher, D. V. (1996). A farewell challenge. School Library Media Quarterly 24, no. 4, pp. 192-194.

Montgomery, P. (1997). *Use of information*: The information Literacy phenomenon. In Tepe, A. E. & Calarco, J. (Eds.), A handbook for pathways to Knowledge (pp. 8-10), Follet's Information Skills Model. McHenry, IL: Follet Software.

Thomas, N. P. (2004). *Information literacy and information skills instruction: Applying Research to Practice in the School Library Media Center*. London: Libraries Unlimited Inc.

CHAPTER TWO

LIBRARIES

By

Juliana Amavi Opare-Adzobu, Paulina Afful-Arthur, Korklu A. Laryea and Christopher Kwame Filson

2.0 Introduction

People acquire education through certain institutions like schools, churches, welfare bodies, associations and organisations and the library is the most outstanding of such institutions. A library may be defined as a building or a series of rooms containing collections of records of human culture in diverse formats and languages, preserved, organized and interpreted to meet broad and varying needs of individuals for information, knowledge, recreation and aesthetic enjoyment (Alemna, 2000). Aina (2004) sees libraries as institutions responsible for the collection, processing and storage of recorded knowledge for the purpose of reading, study and consultation.

Librarians therefore acquire, catalogue, classify and arrange the library materials in such a way as to facilitate easy retrieval. The specific activities involved are as follows, and are done sequentially in various departments:

- Acquisitions – involve the selection of the materials based on the information requirements of the specific library and the needs of the clientele. Materials may be purchased, exchanged

with other libraries or donated by institutions and individuals to stock the library's collection;

- Processing of the materials involve stamping, cataloguing, classifying, and lettering. – This enables each material to be associated with a particular library. Each item acquired is described and designated to the appropriate subject area(s);
- Displaying the materials in the appropriate areas in the library either for borrowing or reference by users under the Lending or Reference Sections.

2.1 Types of Libraries

There are different types of libraries and they are set up to serve a specific group of people. Your own collection of books is your personal library. However, those set up by institutions and governments are of five main types – academic, special, school, public and national. These types in turn are classified under three broad forms – traditional, electronic/digital, and hybrid - based on how the activities in the preceding paragraph are performed. The library serves as a place for pupils, students, researchers and individuals desiring to study, or to do independent work/research. Libraries have demonstrated that when well utilized they have a positive impact on users' lives and academic achievement.

2.1.1 School Libraries

A school library can be defined as the library within a pre-tertiary institution where pupils/students and staff of a public or private school have access to a variety of resources. Morris as quoted in Wikipedia also asserted that a school library serves as the centre

and coordinating agency for all materials used in the school (http://en.wikipedia.org.wiki/school_library, 25th July, 2013). The main function of the school library is to support the various educational programmes at the various levels. The school library exists to provide a range of learning opportunities for both large and small groups as well as individuals with focus on intellectual content, information literacy, and the learner (Morris, 2004).

Very few schools in Ghana at pre-tertiary level have libraries to boast of. Some of them have limited collections. However, some private schools (eg. Ghana International School and Akosombo International School) have developed comprehensive hybrid collections of traditional print and non-print resources, and employ qualified librarians to manage their libraries. Schools in the rural areas are even less likely to have any room or collection to call a library.

Collections

The collections of school libraries may consist of printed works such as general reference books like dictionaries, encyclopaedias, atlases, and periodicals (such as magazines and newspapers), and those that can be taken on loan such as textbooks and novels. Non-print materials may include films, audio and video recordings, computers, and CD-ROMs. A good school library may also have television set for use. The content of the library materials depends on the various educational levels (i.e. primary or secondary).

Services:

Reference
The School Librarian assists users to find materials in the collection.

Lending
This service affords users the opportunity to borrow books to use outside the library. Novels which may be fiction and non-fiction can therefore be borrowed by students and used at their leisure and returned at the end of the lending period.

In addition, well endowed school libraries can offer electronic and reprographic services.

2.1.2 Academic Libraries

An academic library is also known as a college or university library. It serves the curricular needs of students, faculty and staff of the parent institution (ie. post-secondary institution). In other words, its users are the academic staff (lecturers), researchers, students (both under graduate and post-graduate) and administrative staff.

Collections
The main function of the academic library is to support the teaching, learning and research activities of the institution. The academic library is often described as the nerve-centre of educational institutions especially the universities. Every tertiary

institution is required to have a library to enable it to offer academic programmes. Among the fundamental criteria for giving accreditation for academic programmes is the adequacy of library collections to support the curricula. The library therefore develops a collection of resources based on all subjects being taught in the university. Consequently, the collection reflects the academic programmes offered by the parent institution. As new programmes are established and existing ones expand into new areas, the library collections are expected to expand proportionally. These library resources include general reference books, textbooks, periodicals, newspapers, government publications in the print format and a wide range of electronic information including access to the Internet, full-text databases on CD-ROM, e-books, microforms, etc. since most academic libraries are hybrid in character.

The collection in the library is most likely to be either for reference purposes or for lending. Materials tagged for reference can only be used on the library's premise. Those designated as lending can be borrowed or loaned out for a period of time.

It is important to note that apart from the main academic library of the institution, there could be branch libraries like hall, faculty and departmental libraries. For instance, in the University of Cape Coast, the Department of Basic Education has a Children's Library which is used by students offering Basic Education. Children in the basic schools of the University also use this library

Services

Reference

Under this service, the librarian/qualified library staff serves as a link between the user in search of information and the needed resource. The library user's questions can be answered at the reference desk, or with the aid of a catalogue, in person, via the telephone or electronic mail.

Lending

The largest collection of books in most university libraries are available for lending. Borrowing can vary from a few hours, overnight, days and sometimes weeks. This service enables users to borrow materials for use outside the library. Users are required to pay for lost books and pay fines for overdue books. They can also ask for books to be reserved for them.

Inter-library Lending & Document Delivery (ILL/DD)

No matter how well endowed libraries the world over are, they cannot provide on-site all the needs of their patrons. Libraries therefore have agreements to borrow materials from other libraries on behalf of their patrons. The two services are part of resource sharing or library cooperation.

Inter-library lending is a transaction in which the user makes a request with his local library which, acting as an intermediary, identifies owners of the desired item, places the request, receives the item, makes it available to the user and arranges for its return.

Most frequently, requested items include theses, rare books, and textbooks.

Document delivery, however is the provision of non-returnable documents, published or unpublished, in which ever format, at an established cost upon request by users (Baker and Jackson, 1993; Alemna, 1997). Materials for this service are usually duplicated and sent by electronic or snail mail to the recipient.

Electronic Support

This deals with the use of resources in electronic format and includes databases on CD-ROM, and online databases subscribed to by libraries. These resources are part of the reference collection because they can be accessed from terminals or computers located within and outside the library.

Reprographic

This service includes the reproduction of materials for use by patrons. Materials can be photocopied or with the use of digital duplicating devices (scanners or cameras) to produce an exact copy of existing document(s) for a fee at the request of users. One would use this service to make copies of materials within copyright and fair use policies. It is therefore a breach of the law to make copies of a whole book. One would normally use this service for reference materials that cannot be borrowed or are too fragile to be copied. Such documents sometimes attract higher fees when duplicated.

User Education

This service can be in two tiers. In the first tier, the library organizes User Education for fresh students through library orientation including a guided tour of the library, to enable them to use the library effectively. The second tier of this service is the introduction of compulsory Information Literacy course in some universities. The following are some of the advantages of this service:

- Knowing how to use the library makes students independent learners and this promotes efficiency in individual study, research and intellectual self reliance.
- One needs the library to perform such tasks as assignments, project works, research activities because the library provides materials which cut across all subject areas;
- One gains ability and skills to search for information.

2.1.3 Special Libraries

This type of library is found in government institutions, industrial, research, commercial and business organisations to provide for the information needs of their staff. The overriding requirement of the special library is the provision of information that enables research workers and other employees to carry out their company duties affectively.

One of the oldest special libraries in the country dates back to the 1890s with the establishment of the Aburi Botanical Gardens (Alemna, 2000). Its primary purpose was to support the exploration of economic plants and other agricultural resources

for European markets. It also supported the experimentation of ornamental plants from other tropical and sub-tropical climates to thrive in our local environment. Its collection was absorbed into the Ministry of Food and Agriculture library which sadly has lost its lustre.

Collections

The collections of special libraries include general reference materials, relevant books and periodicals both in print and non print formats. The contents of the reference sections of special libraries vary according to the needs of their particular organisations.

In the academic setting, the Johannes Zimmermann Library of the Akrofi-Christaller Institute of Theology, Mission and Culture in Akropong-Akuapem for example is an example of a special library. Its collection consists primarily of materials in theology, religion, mission, and Ghana. It is predominantly for research and thus provides limited lending service for users and reprographic services for walk-in users.

The Institute for Scientific and Technological Information (INSTI) serves as a central library that coordinates the activities of all the special libraries in the research institutes of the Council for Scientific and Industrial Research (CSIR). It is a documentation centre and has developed an efficient method of recording and retrieving information. The CSIR group of libraries has been developed around the various research disciplines to

serve the information needs of the research scientists. These include the Animal Research Institute library, Food Research Institute library, Oil Palm Research Institute library, Water Research Institute library, Crops Research Institute library, etc.

Another example of a special library is the Supreme Court of Ghana Library which was established in 1876 with the establishment of the Supreme Court of the Gold Coast. Its collections include legal documents (statutes, law reports, reports of commissions of enquiry, bulletins, cases,) produced by the various courts of the land, the United Kingdom and the British Commonwealth. A law firm on the other hand may maintain its own library of legal documents for its lawyers and staff and depend on the Supreme Court Library for more historical and other legal materials. Other examples of special libraries are at the Geological Survey Department, Lands Department, Meteorological Services Department, Ministry of Information and the Statistical Service of Ghana.

Services

Special libraries are often set up as reference libraries and provide services similar to those at academic libraries (reference and reprographic services). They however include additional services (listed below) which are peculiar to them:

Selective Dissemination of Information (SDI) which is a customized service and very common in most special libraries. The library users are updated on new publications related to their field of study or research interest. The library staff on regular

basis search through a mass of documents from which those that are of interest to a particular user are selected. The user of course would have to provide the library staff with their research or study profile. This service relieves the user of the problem of sieving through a large number of documents for relevant information;

Abstracting Service - whereby library staff provide added value to the document being sought by users by providing the full bibliographic details and a summary of the document;

Indexing Service is the process of providing a guide to the bibliographic details of a document so that information can be retrieved easily;

Reference service – especially for walk-in users.

2.1.4 Public Libraries

Public libraries by their nature are the most familiar and visible type of library to most people. A public library is a library that has been set up to provide service to all members of the community regardless of age, citizenship, occupation, economic status, educational level etc. Public libraries are funded by government and are sited in cities, towns or villages so that users can have easy access to their facilities. Public libraries are distinct from research libraries, school libraries and other special libraries in that their mandate is to serve the general public's information needs. Community, municipal and district libraries are all examples of a public library.

Ghana's first attempt at providing public library service was in 1928 by Reverend John Anglioby, an Anglican Bishop in Accra. In 1950, the Ghana Library Board (now the Ghana Library Authority) was established and mandated to establish, equip, manage and maintain all public libraries in the country. There are ten regional libraries and several municipal, community and district libraries that are supervised by the Ghana Library Authority.

Collections

Public library collections typically include popular reading materials for children and adults, basic reference sources, classics and resources specific to the local economy and culture. Also included are general textbooks, storybooks, encyclopedias, dictionaries, newspapers and other materials (print and non-print).

Services

Public libraries serve diverse constituencies which include children, the aged or home-bound, students, adult learners, the visually impaired and physically challenged, and prisoners in their communities and therefore provide a wide range of services. These include:

- Reference Service
- Lending Service
- Reprographic Service

- Extension Library Service (includes mobile services to prisons and hospitals).
- Story telling sessions for the infants, toddlers, preschool children, or after school programmes to encourage early literacy;
- Some public libraries provide art and craft materials for children, often housed in a special section. In the developed countries, child oriented websites with on-line educational games and programmes designed for younger library clients are becoming popular. (http://en.wikipedia.org/wiki/public_library);
- Provision of community meeting rooms and study areas for quiet study for students and working professionals; and
- Support for Book Clubs to encourage appreciation of literature and reading in adults.

In Ghana, the Ghana Library Authority, with its headquarters in Accra is responsible for providing public library services in the country, to meet the information needs of various communities.

2.1.5 National Library

A national library is a library specifically established by the government of a country to serve as an outstanding repository of information for that country. A national library is usually the largest and the most comprehensive library in its coverage of information about a country.

The main goal of a national library is to ensure the bibliographic control of all books and other materials published in that particular country and about that country.

Collection

The collection includes all published works in a particular country and about that country, both in print and non-print format including electronic resources. These include books, newspapers, magazines, journals, government publications reflecting in a way the publishing output of the country.

Functions

- Legal deposit (the national library is supposed to receive free of charge at least one copy of every work published in the country);
- Compilation of the national bibliography;
- Storage of numerous valuable works and rare books;
- Practical training for library personnel;
- Conducting research as a consultancy service for individuals and the nation.

Services

- General reference service
- Inter-library Lending and Document Delivery at national and international levels

Examples of national libraries are the British Library (UK), the Library of Congress (USA), the National Library of the Republic of South Africa (RSA).

Ghana unfortunately does not have a national library, however, the George Padmore Research Library on African Affairs serves as the national library. Apart from being an Africana library with special emphasis on Ghana, the library performs the following essential functions of a national library for the country:

- Acquisition, processing, preservation, documentation, accumulation of domestic and Africana publications and providing access to them;
- It is the national bibliographic agency and a legal deposit library (copyright law);
- Management of the International Standard Book Number (ISBN) and International Standard Serial Number (ISSN) systems;
- Guidance and support for library services and learning activities for other libraries;
- Cooperation with foreign libraries and international exchange of materials;
- Compilation of special bibliographies as a means for research. Retrieved fromhttp://www.people.tamu.edu/-yarak/padmore_request.html on 30th August, 2013

2.2 Forms of Libraries

2.2.1 Traditional Library

In a traditional library, most of its operations (activities) are done manually and without the aid of computer technology. In addition, almost all the library's resources are in print format. Non-print materials (audio-visual materials such as audio and video tapes, microform) are accessed with the appropriate machines.

Advantages of Traditional Libraries

The following are some of the advantages:

- Stable collection;
- Free and universal access to the collection;
- Can be used without electricity;
- Documents can easily be photocopied;
- No computer expertise is required on the part of the library staff and users.

Disadvantages of Traditional Libraries

The following are some of the disadvantages:

- Limited access to materials;
- Centralized management of the resources;
- Users are limited by physical location;
- Content of materials are not directly linked with each other;
- Slow update of collection.

2.2.2 Electronic/Digital Library

A digital library has all information resources in digital form and access to such resources is through the use of digital technologies. It is therefore a library in which collections are stored in digital formats (as opposed to print, microform, or other media) and accessible by computers.

Electronic devices such as digital cameras and scanners are used to digitize printed specific library collections. Such materials are made-digital. Other materials are born-digital in that they were made with computer technology and need not be converted into digital format. Digital libraries can be accessed on the internet.

To be considered as a digital library, an on-line collection of information must be managed and made accessible to a community of users. The digital content can be stored locally or remotely and can be accessed via computer networks.

Many academic libraries are actively involved in building institutional repositories made up of the institution's books, papers, theses and other works which can be digitized or are born-digital. Examples of such institutions in Ghana are Kwame Nkrumah University of Science and Technology, Kumasi, University of Cape Coast, Cape Coast, University of Education, Winneba and University of Ghana, Legon.

Advantages of Digital Libraries

The following are some of the advantages:

- They allow users to directly access and use the documents information they have found;
- Access to information is quicker because of the use of computers and a network of computers. They allow users to quickly find documents by using a computer and a network;
- Information can be accessed anywhere and at any time (there are no closing hours) since one does not need to be physically present in the holding library. ;
- Digital library collections can be delivered on CD-ROM and other external storage devices to users with inadequate network connectivity;
- A digital library can provide access to content in different and more appealing forms including animation, graphical, audio and video formats;
- Information is more easily shared with other digital libraries. There is therefore enhanced access to users.

Disadvantages of Digital Libraries

The following are some of the disadvantages:

- The content of some of the documents may not be well screened and validated;
- Difficulty of knowing and locating everything that is available and differentiating valuable from useless information if one is not computer literate;
- Job loss for traditional publishers;

- It cannot be used when there is no electricity and inadequate bandwidth;
- It is more expensive to set up and manage.

2.2.3 Hybrid Library

The hybrid library has its information resources in both print and non-print formats. It also means that some of its operations or services (such as acquisitions, cataloguing and circulation) are performed with the aid of computer technology. This implies that the users of a hybrid library have access to physical collections as well as digital collections (electronic resources) like full text on-line journals, e-books, audio books and full text databases on CD - ROM etc. In such libraries for example, online public access catalogues (OPACs) are used to access the libraries' collections. All the features of the traditional library may be present in the hybrid library.

Most academic libraries develop hybrid collections of print and electronic resources and provide integrated access to them. With a common gateway members of the university community can use computers in the library, in offices, in halls or elsewhere to identify what resources are owned by their institution, what is on order, and the physical materials currently are on loan to other borrowers. Some systems enable users to request for items for delivery to their offices and extend the due dates for borrowed items. Most academic libraries are of this type whereas most of the departmental libraries are of the traditional type since the

services and operations do not employ/use technological interventions.

Advantage of Hybrid Libraries

The main advantage of this form of library is that information can be accessed in both print and non- print formats. It harnesses the advantages of both the traditional and digital libraries.

References

Aina, L.O. (2004). *Library and information science text for Africa*. Ibadan: Third World Information Services Limited.

Borgman, C.L. (2000). *From Gutenberg to the global information infrastructure*. Cambridge (MA): MIT Press.

Gates, J.K. (1962),*Guide to the use of books and libraries*. New York: McGraw-Hill Book Company.
http://www.people.tamu.edu/-yarak/padmore_request.html retrieved on 30th August, 2013

Mayank, T. (2010). Digital libraries: functionality, usability, and accessibility. Library Philosophy and Practice (e- journal). Retrieved from http://digitalcommons.edu/libphilprac/381 on 23rd August, 2013.

Morris, B. J. (2004). *Administering the school library media center*. Westport CT: Libraries unlimited.

CHAPTER THREE

INFORMATION SOURCES AND THEIR USES

By

Korklu A. Laryea, Christopher Kwame Filson, Paulina Afful-Arthur and Juliana Amavi Opare-Adzobu

3.0 Introduction

Resources in libraries include general reference books (like encyclopaedias, dictionaries, atlases, directories), textbooks (on various subject areas), novels, periodicals (journals, newsletters, bulletins, newspapers, etc.), and electronic resources (which may be on compact disc (CD) or other external storage devices or online). These resources may be print and non-print materials. Increasingly, technology is making it possible for reference and other materials to be in electronic format. This chapter focuses mainly on printed reference sources and their uses. Examples are given of the various materials discussed and effort must be made to examine and evaluate as many examples of materials for their effective and appropriate use during tutorial sessions.

Library reference resources are essentially grouped into two:

- Directional Search tools which direct or point the reader to other materials where information can be found. They do not contain the actual information. Examples are bibliographies, indexes, and abstracts.

- Source Type resources which contain the information needed. Examples of these are dictionaries, encyclopaedias, journals, and thesauri.

3.1 Reference Materials

Reference materials or reference sources contain facts that have been brought together from many sources and organized for quick and easy use. According to Aina (2004), reference sources are documents that contain information on any topic – be it an event or a person. For the purposes of this course only a few examples will be selected for study.

Arrangement of information in reference books/sources may be:

- Alphabetical (as in the case of dictionaries and encyclopaedias);
- Tabular as found in mathematical and statistical works;
- Regional such as maps and atlases;
- Chronological (as in the case of the Bible and history books);
- Classified - according to some features or characteristics (as in bibliographies and directories).

3.1.1 Characteristics of Reference Materials

- They are meant to be consulted in the library. Such books are likely to have 'Reference Only', 'Not To Be Taken Out' or 'Not To Be Borrowed' labels on them;
- They are not meant to be read from cover to cover;
- They are housed in separate sections of the library such as the Reference Section;

- They are revised from time to time so as to keep abreast with recent developments;
- A library usually buys a copy or in rare cases two or a set of one or two because they tend to be expensive.

Though some of these materials may be electronic and therefore offer unimpeded access, some need to be subscribed to for accessibility for various reasons (such as security and economic reasons). Scholarly electronic journals, for example are subscribed to by individual libraries or through a consortium such as the Consortium of Academic and Research Libraries in Ghana (CARLIGH) and the South African National Library and Information Consortium.

3.1.2 Evaluating a Reference Material

Reference materials are prepared for specific audiences for specific purposes. One needs to therefore assess each reference work according to one's information needs. The following assessment points or principles are applicable in evaluating every reference work or publication and are often stated in the publication's preface or introduction. For example, when assessing a dictionary, its authority, accuracy and currency are essential, since dictionaries tend to be prepared for a specific audience (age group, students of a subject area) or a specific purpose. Therefore it is essential to evaluate the degree to which a dictionary has succeeded in fulfilling its purpose. The format and ease of use are as helpful in evaluating reference materials as are the following:

- Authoritativeness of the author(s), editor(s) or contributor(s) qualifications. The reputation of the publisher is also important;
- The date and edition of the publication;
- The scope of the work – the preface introduces the author(s) or publication to the reader and gives reasons for writing the book, the intended audience and explains the arrangement, symbols and abbreviations as well as any special feature(s) used. The scope of the material includes the geographical, language, and period/time;
- Comprehensiveness in relation to depth or maximum coverage of the topic(s) or issue(s) discussed in relation to the intended audience;
- Accuracy refers to the authenticity of the facts relevant to the subject.

3.2 Dictionaries

A dictionary is defined as a reference book or a set of books containing words and usually phrases, alphabetically arranged, together with information about them, especially their forms, pronunciations, usage, spelling and meaning. In addition, the dictionary may give etymology, synonyms, antonyms, biographical facts and geographical information or gazette.

3.2.1 Types of Dictionaries

- General dictionary (single language dictionaries) such as *Webster's Third New International Dictionary* (Springfield: Merriam-Webster, 2002) covers all aspects of words in all

fields of knowledge, *Oxford Dictionary of Englsih*, 3rd ed. (Oxford: OUP, 2010), *Shorter Oxford English Dictionary*, 6th ed., (Oxford: OUP, 2007) is in two volumes; *Colins COBUILD Advanced Dictionary* (Glasgow: Harper Collins, 2009);

- Translating dictionaries are bilingual dictionaries or polyglot dictionaries (where the words in one language are translated into two or more languages). Examples are *An English, Akan, Eve and Ga Dictionary* compiled by C. A. Akrofi, G. L. Botchwey and B. K. Takyi (Accra: Waterville, 1968); *A Standard Englsih-Swahili Dictionary*, (Oxford: OUP East and Central Africa, 1992), *Swahili-English, English-Swahili Pratical Dictionary* by Nicholas Awde (NY: Hippocrene Books, 2000);
- Subject dictionaries – every academic discipline has its own set of lexicon, terms and vocabulary of words and phrases peculiar to it. The words 'mouse' and 'boot' mean different things in the English language and information technology. Good subject dictionaries in addition to defining words, give explanations of their meanings, illustrate their use and are cross-referenced. Examples of such publications include *Chambers Dictionary of Science and Technology* (Edinburgh: Chambers, 1999); the *Oxford Concise Companion to Classical Literature* (Oxford University Press, 1993); the *Longman Dictionary of Contemporary English*, 3rd ed. (Edinburgh: Pearson International Ltd., 2001) includes a CD-ROM which is actually the electronic version of the dictionary; *Dictionary of Information Technology*, (New

Delhi: Firewal Media, 2010) by Ramesh Bangia; *Biotech Dictionary of Pesticides*, complied by Prateek Mittal, Indian ed. (Delhi: Biotech Books, 2007); *The Encylopedic Dictionary of Environmental Change*, edited by John A. Matthews (London: Arnold, 2003);

- A specialized dictionary deals with certain aspects of words such as slangs, abbreviations, synonyms, antonyms, and usage. Examples include *Collins Roget's International Thesaurus* (Glasgow: Harper Collins. 1996); *Bloomsbury Spelling Dictionary* (London: Bloomsbury Publishing, 1994); *Cambridge International Dictionary of Idioms* (Cambridge University Press, 1999); *Using French Synonyms* by R. E. Bachelor and M. H. Offord (Cambridge University Press, 1993); *Macmillan English Dictionary for Advanced Learners* (London: Macmillan Education, 2002); *A Dictionary of Modern English Usage* by H. Fowler, 2nd ed. (Oxford: OUP, 2003).

3.2.2 Uses of Dictionaries

Dictionaries are used to define words, to verify spelling, syllabication or pronunciation; to check on usage; or to determine the etymology of a word. To some extent, they also standardise the language based on current usage.

3.3 Encyclopaedias

Reitz (2013) defines an encylopaedia "as a book or numbered set of books containing authoritative summary of information about a variety of topics in the form of short essays, usually arranged

alphabetically by headword or classified in some manner". The entries may have the name(s) of contributor(s), be illustrated, and have a list of references for further reading at the end of the text. Multiple-volume encyclopaedias have indexes usually in the last volume to facilitate the location of information in the entire publication/set. For Aina (2004), an encyclopaedia is a reference source devoted mainly to providing information and facts on a variety of topics. The articles are written by experts in the field and the topics are arranged alphabetically.

The content of an encyclopaedia usually gives an overview of a topic including definition, description, background and bibliographical references. The most important of reference books are probably encyclopaedias as they contain information about places, events, people and things, including facts and figures.

3.3.1 Types of Encyclopaedias

- General encyclopaedias – The *Webster's New Explorer Desk Encyclopedia* (Springfield: Federal Street Press, 2003) is a single-volume encyclopaedia. An example of a multiple-volume encyclopaedia is the fifteenth edition of the *New Encyclopaedia Britannica* which is regarded as the most scholarly. The current edition (2010) is in thirty-two volumes and consists of three parts: the **Propaedia** which provides an outline of knowledge covered, the **Micropaedia** (in twelve volumes) provides basic information, and the **Macropaedia** (in nineteen volumes) gives more detailed information. It also has a two-volume index;

- Encyclopaedia for different age groups (example for children and young adults); *The New Children's Encyclopedia* (DK Publishing, 2013); *Scholastic Children's Encyclopedia* (Scholastic Reference, 2004);
- Subject encyclopaedias as implied have subject coverage or focus on particular subjects. *The Encyclopedia of the Social Sciences* (New York: Macmillan, 1967) was in fifteen volumes and had sixteen editions; the seventeen-volume *International Encyclopedia of the Social Sciences* (New York: Macmillan, 2008) has replaced the *Encyclopedia of the Social Sciences; Black's Medical Dictionary* (London: A&C Black, 2010) is in its 42nd edition; Geoffrey Parrinder, *A Concise Encyclopaedia of Christianity* (Oxford: One World, 1998);
- Encyclopaedia yearbooks and supplements –*Africa South of the Sahara* 2011, 40th ed. (London: Routledge, 2010) is a worthy publication for anyone interested in the geographical area under scrutiny.

3.3.2 Uses of Encyclopaedias

- Finding facts;
- Provide general background or historical information;
- Locate explanatory material or definitions;
- Serve as guides to related topics within its pages or to outside sources of information (cross references and bibliographies);
- Teach research skills, that is, a systematic approach to gathering information.

Like dictionaries, all the above publications may be in single or multiple volumes.

3.4 Directories

Reitz (2013) describes a directory as a list of people, companies, institutions, organisations, etc., in alphabetical or classified order providing contact information (names, addresses -physical, postal, email, website, phone and fax numbers) and other pertinent details (functions, affiliations and membership of associations, conferences, publications, etc.)

3.4.1 Uses of Directories

Directories are used to locate organisations, institutions and people. They are also used to verify the name of an organisation or the spelling of a person's name. Directories, like bibliographies and other databases are to enable users make the best use of and provide access to information about people, organisations, institutions as well as publications.

3.4.2 Types of Directories

Like the other reference materials above, there are different types of directories to serve different needs. These are

- Telephone and facsimile directories are the most common and contain the names, addresses and telephone numbers of organisations, institutions, associations, and individuals. They are excellent primary sources for local information;
- Institutional or organisational directories which are valuable sources for identifying the location, contact details, staff and

areas of specialization, support or service. This may cover educational, research, commercial or business enterprises. *The World of Learning,* 39th ed. (London: Europa Publications, 1989); *Encyclopaedia of Associations* (London: Fredrick C. Ruffner, 1973) *World List of Universities and Other Institutions of Higher Education*, 23rd ed. (Paris: IAU/UNESCO Information Centre on Higher Education. 2002); *The ABS Directory of Business Schools* (Oxford: Butterworth-Heinemann Ltd., 1997); *Directory of Libraries and Library Personnel in Ghana* (Accra: Ghana Library Association, 2004);

- Publishing and book trade directories provide the basic contact information about organisations and companies involved in buying, selling, or exchanging a specific category of goods and services;
- Directory of directories – provide listing and description of the various directories. They are a good starting point when a user needs a specific directory but does not know the exact title;
- The *Directory of Open Access Journals* http://www.doaj.org/ is an example of an online directory of electronic journals.

3.5 Bibliographies

The word bibliography comes from the Greek word *bibliographia* meaning book writing. A bibliography is the name given to a list of books, manuscripts and other publications systematically arranged and described and which have some relationship to each other. According to Bopp (1991), bibliographies are central to the

reference and information processes in that they guide the information user in the 'chaotic' world of books and other forms of information.

3.5.1 Types of Bibliographies

- **Author** bibliographies list the works by and about one author. They may be found at the end of the author's publication (Kwame Bediako, *Jesus in Africa: the Christian Gospel in African History and Experience*, Akropong-Akuapem: Regnum Africa, 2000; Mercy AmbaOduyoye, *Beads and Strands: Reflections of an African Woman on Christianity in Africa*, Maryknoll: Orbis Books, 2004) or part of the curriculum vitae (CV). They may be obtained electronically on the person's personal website or blog, professional and academic websites, or electronic social media (eg. on Facebook or LinkedIn);
- **Subject** bibliographies are restricted to a specific subject area. E.Y. Amedekey, *The Culture of Ghana: A Bibliography* (Accra: Ghana Universities Press, 1970); Dennis M. Warren, *Bibliography and Vocabulary of the Akan (Twi-Fante) Language of Ghana* (Bloomington: Indiana University Press, 1976); Patrick E. Ofori, *Christianity in Tropical Africa: A Selective Annotated Bibliography* (New York: Kraus International Publications, 1977);
- **The National** bibliography includes all materials published in a country or about a country in a given year. In essence, the national bibliography is the intellectual output of a country, for example, the *Ghana National Bibliography* (GNB). The

most popular national bibliography is the *British National Bibliography* (BNB) which lists the books and new journal titles published or distributed in the United Kingdom and Ireland since 1950. It also lists forthcoming book titles and hand-held electronic publications (eg. CD-ROMs), deposited with the Legal Deposit Office since 2003;

- A **Book Trade** bibliography, whose basic function is to provide information about printed works that are in stock. They may also include recently published or about to be published works, to facilitate the acquisition of publications. They may be compiled by a publisher (Oxford University Press general catalogue, Orbis Books general catalogue) or book vendor (*Whitaker's Books in Print*, and *British Books in Print*);
- Publishers' catalogues or book trade bibliographies are now obtainable electronically from the publishers' websites. Bibliographies may also be found in individual books, periodical articles, encyclopaedias and other reference books. These citations can be followed to get more information on one's area(s) of interest or research.

3.5.2 Uses of Bibliographies

- Bibliographies are useful sources in searching for materials/publications on a subject or person/author. That is, they can be used to locate materials on the subject in question or about the number of publications emanating from an author;
- They provide a means to verify the details of publications;

- They indicate the scope of the subject;
- The bibliographies found in articles and books provide further information for users to investigate their areas of research and study.

3.6 Biographies

A biography is the story of a person's life written by someone else. Biographic reference sources are special reference sources that provide information about individuals whether dead or alive. The basic information in biographical reference materials include the name of a person, date and place of birth, background (family, country, educational, professional, etc.), and his or her role in society.

3.6.1 Types of Biographies

According to Aina (2004), the three major categories of biographical sources are

- **Universal** biographies are usually unrestricted in their coverage and contain the life history of achievers the world over. The 58th edition of *The International Who's Who 1994-1995* (London: Europa Publications) was published in 1994;
- **Regional** biographies include people who have played significant roles in a region (within national and international boundaries). D.E.K. Amenumey, *Outstanding Ewes of the 20th Century: Profiles of Fifteen Firsts, volume 1* (Accra: Woeli Publishing Services, 2002); *Dictionary of African Biography* complied by Ernest Kay (London: Melrose Press,

Ltd., 1970), *Makers of Modern Africa* (3rd ed., London: Europa Publishing, 1996) and *Africa Who is Who*, 3rd ed. (London: Europa Publishing, 1996) are regional in content in that they include people from all over the continent;

- **Professional or Subject** biographies are restricted to those that are involved in a particular profession, discipline or subject area. An example is *The Cambridge Dictionary of Scientists*, 2nd ed. (Cambridge: Cambridge University Press, 2002) compiled by David Millar, Ian Millar, John Millar and Margaret Millar because of their interest in science but it is not universal in scope. *The Dictionary of African Christian Biography* is an electronic compilation of significant Africans in Christendom from earliest times. It is a dynamic project with world-wide contributors whose inputs undergo peer reviews before being posted on the website (http://www.dacb.org).

3.6.2 Uses of Biographies

- Find information about a person;
- Locate people who are famous in a given country, occupation, career or profession;
- Provide useful historical information and therefore enriches history;
- Prepare towards introducing a guest speaker (if it is current).

One should be mindful that these publications are in bound volumes. However, biographies may also be found in other reference/periodical publications such as yearbooks, facts on file,

and periodicals (journals, newspapers and magazines) and thus will not be in a bound volume. Oral recordings of interviews on radio and television are also good biographical sources. *Obaa Mbo* and *Egya Pa* (on TV Africa), *Personality Profile* (on Joy FM and the Joy News Channel on Multi TV) as well as *Senior Citizen's Corner* which has been aired on GBC Radio Two for several years.

3.7 Geographical Sources

Geographical information sources are those in cartography. Maps and atlases are important in educational and business activities as well as in everyday life. Geographical sources are used to answer a range of enquiries from a simple request such as finding the best and quickest route from one point to the other, to complex academic and planning problems. For example, Ghana's geographical location on the Greenwich Meridian could be advantageous in the world of commerce.

3.7.1 Maps and Atlases

A simple definition of a map is a representation of the surface of the earth or a section of it. An atlas is a bound collection of maps, and may include charts, plates or tables, illustrating any subject. There are various types of maps including, topographical, political, road, economic/resource, and climatic in various formats (printed or electronic) used for different purposes and information needs. Maps are usually stored flat in special cabinets and require large surface areas to be used. They are therefore likely to be kept

separately. Maps or atlases fall into two major groups: thematic and topographical.

Topographical maps are representations of the land surface. Thematic maps on the other hand are representations of particular/specific features or spatial distribution. Maps of this nature may include the following: geological maps, land-use maps, road maps and charts (used by navigators).

Traditionally, atlases could be defined as collections of maps of uniform dimension bound together in a single volume or volumes. They may also be found in loose leaf format or be available electronically. Atlases may be the work of individuals, publishers (Philip's, Times or Oxford) or learned organisations (eg. National Geographic Society) Atlases may therefore include maps which are topographical and thematic (tourist, religious, medical) in nature. There are national atlases which are collections of both topographical and thematic presentations of a particular country often derived from official sources. *The Philip's World Atlas*, Ref. ed. (London: Chancellor Press, 1995) is more than a compilation of maps to include a gazetteer of countries (giving basic information about each country).

3.7.2 Types of Atlases

- General reference atlas - *The Times Atlas of the World* (London : Times Books, 2005), *World Bank Atlas* (Washington, D.C.: World Bank, c2004) ;

- Regional atlases - covering nations/countries, sub-regions and continents *Regional Atlas on West Africa* (Paris: OECP, 2009), *Atlas of the Gulf states* (Boston: Brill, 2013);
- Road atlases – *Road Maps of Kuala Lumpur and its Surroundings* (Kuala Lumpur: Upa Press Sdn. 1986); *Israel: the New Road Atlas* (Tel Aviv: MAP-Mapping and Publishing, 1994), *Singapore Street Atlas: Includes Johor Bahru* 2005-2006 (Singapore: Periplus Editions, 2004);
- Historical atlases – *The Atlas of the Later Zulu Wars*, 1883-1888 (Pietermaritzburg: University of Natal, 2001); *A missionary atlas, for use in connection with mission study circles, etc.: maps of Fiji, New Britain, Papua, the South Pacific, Solomon Islands, India, Goulburn Island (N.T.)*[Sidney]: Methodist Missionary Society of Australasia, [1921];
- Thematic atlases – *The Moody Atlas of the Bible* (Chicago: Moody Publishers, 2009), *Philip's Atlas of the Oceans* (London: George Philip, 2003); *Atlas of Human Anatomy* (Philadelphia: Saunders, 2011; *An Atlas of the Surgery of the Spine* (London: Martin Dunitz, 1998); *Cultural Atlas of Africa* (Oxford: Phaidon, 1981).

3.7.3 Uses of Atlases

- Locate places on the earth's surface;
- Determine the distance between two points;
- Determine the time zones of cities and countries for decision-making.

3.7.4 Gazetteers

The gazetteer is an alphabetical arrangement of place names, including both natural and man-made features and is often used alongside maps and atlases. The word *gazetteer* was originally used to describe a writer for a gazette (an official newspaper report). Gazetteers may be found in the last pages of an atlas and give some reference to the location of the features and other additional information of places listed/mentioned. They may be considered as the index of the atlas.

According to Katz (1997), the gazetteer is the reference book first used in attempting to locate a place. The gazetteer is basically a listing of places with sufficient information to identify their location. Most gazetteers add more information about a place. As with all reference works the gazetteer providing international coverage is less comprehensive and probably less informative about the places within one country than a gazetteer devoted to that country alone. The gazetteer listing is arranged alphabetically and is therefore dependent on the orthography used.

3.7.4.1 Uses of Gazetteers

- Locate places;
- Check the approved names and places;
- Check the correct spelling of place names.

3.7.5 Travel Guides

They are documents meant to provide information to the traveler about a place of interest. The increase in tourism activities has

generated an unprecedented proliferation of travel literature, both domestic and international. The Ghana Tourism Authority has published a number of travel guides (eg. *Ghana Today*) which list and describe the hotels, restaurants, tourist centres (such as game reserves, forts, museums and castles, and waterfalls) in the country.

3.8 Periodicals

A periodical is a serial publication with a distinctive title and which appears at regular or irregular intervals, generally more than once per year, without prior decision as to when the last issue will appear. They have assumed greater importance due to their advantage of timeliness over books. Each issue of a periodical is complete in itself but its relationship to preceeding issues is indicated by enumeration. Usually each issue bears a volume, issue number and/or date (eg. *Ghana Library Journal* 21(2), 2009); *The Mirror*, 3058, 17th August, 2013; *African Research & Documentation* 115, 2011.

Periodicals usually contain articles (which may be general or scholarly), stories, photographs, data, advertisements, or writings by several contributors. They include **journals, magazines, reports, newsletters** and **bulletins** whose contents are controlled by an editor or editorial board. These publications have International Standard Serial Numbers (ISSN) made up of eight digits which are unique to them. The frequency of periodicals may range from

- Daily – newspapers fall under this category. Eg., *Daily Graphic, Daily Telegraph,* as with newspapers that are published daily;
- Weekly – as with some newspapers which are published weekly. *The Mirror, Junior Graphic, Weekly Spectator, Weekend Crusading Guide*, etc.
- Monthly - as with some magazines such as *New African, Time, Readers' Digest, National Geographic*;
- Quarterly – which are common with scholarly publications such as *Africa, International Bulletin of Missionary Research, American Anthropology and Journal of African History*;
- Biannual – *Journal of African Christian Thought, General Anthropology*, and *African Journal of Criminology and Justice Studies*

Yearly publications are usually called reports though some reports will appear quarterly or biannually.

Most academic libraries have separate sections for periodicals and are for reference only. The back issues of a particular publication in a year or couple of years are usually bound into volumes and treated like books (catalogued, classified, labeled and shelved). Current issues are displayed face up to make them easily identified and accessible to users. Depending on the design of the display shelf, a number of back issues may be stored under the displayed copies. Back issues of periodicals may be converted to microform or scanned and stored electronically to conserve space. In such instances, appropriate machines (microform reader-

printer) and computers are provided to enable users view and print copies of articles. For many print periodicals, content is also available electronically in full text databases online or on CD-ROM, from commercial vendors or via the publisher's website.

3.8.1 Indexes

An **index** indicates where information can be found rather than provide the information. In other words, an index is a Control-Access-Directional publication, which serves as guide to the contents of a document. Indexes include bibliographies of currently published materials, usually of articles in periodicals, materials appearing in newspapers and literature appearing in collections or anthologies.

The information supplied by indexing publications include the author, the subject of the article or report, the title of the periodical, volume number, date of publication etc. without any added comment(s).

3.8.1.1 Types of Indexes

- **General index** – it is the type of index that covers periodicals in diverse fields with or without geographical bias. For example, *Readers' Guide to Periodical Literature* is a reference guide to recently published articles in periodical magazines and scholarly journals with a geographical bias to the USA and Canada;
- **Subject Index** – very important to scholars and researchers because they provide in-depth of coverage in specific subject areas or topics. The American Theological Library

Association publishes *Religion Index One* (an index to journal articles, essays and book reviews) and *Religion Index Two* (which indexes multi-author works and conference proceedings and festschriften). The two publications can be found in print and electronic formats.

- **Newspaper Index** identifies the date of an event and the corresponding issue of the local newspaper. Example, *New York Times Index* indexes topics that have been published in the *New York Times* newspaper only. A cumulative newspaper index covers different newspapers in a geographical region. The Periodical Section of the University of Cape Coast Library is developing a newspaper index of Ghanaian newspaper the Library subscribes to. This will be available on-line in the near future.
- **Back of Book index** – It is found at the back of books and they are used to find certain topics or words discussed within the text. It could be author indexed or subject indexed or both.

- A **Journal Index** focuses on just one journal title. Eg. *International Bulletin of Missionary Research* is a quarterly journal; and has an author-title index in the last issue of every volume published. Therefore, *IBMR* 36(4), 2012 will have the author-title index for *IBMR* 36.1-4, 2012.

3.8.2 Abstracts

Abstracts give full bibliographic details as in the case of indexes and in addition present a brief summary of the subject content of

the article or report indexed. Abstracts serve as an aid in assessing the content of a document and its potential relevance.

There are several published scholarly indexing and abstracting journals. These are usually located at the Periodicals Section of Libraries (especially academic and special).

3.8.2.1 Types of Abstracts

The two main types of abstracts are; informative abstract and indicative abstract. An **Informative** abstract captures all the essential facts, data, and information in the original document. Scientific new findings are normally abstracted in this form. Informative abstracts are supposed to be surrogates of the original documents. Indicative abstract however, provides an indication of the type of document being abstracted. It is sometimes called descriptive abstract. An **Indicative** abstract shows the principal areas covered by a document and the way the topic has been treated (Aina, 2004). Due to the nature of this abstract it cannot be used in place of the original document because it tells the user what can be found in the document instead of the actual data and information.

3.8.2.2 Examples of Abstracts

As a result of the proliferation of information and the need to provide timely information to users, many abstracts appear as electronic products and are available online via the internet for free or on subscription.

- *Subject abstracts* such as *African Studies Abstracts* is an online service published quarterly by the African Studies Centre (Leiden, the Netherlands) and provides an overview of journal articles and edited works on sub-Saharan Africa in the fields of the social sciences and the humanities. Many of the entries are available in full-text from the Centre's library page - http://www.catalogue.ascleiden.nl. *Ghana Science Abstracts*, published by the Council for Scientific & Industrial Research (CSIR) and Institute for Scientific & Technological Information (INSTI), is an annual compilation of articles published in Ghanaian science and technology journals;
- *Dissertation Abstracts*, published by University Microfilms International (UMI) focuses on post-graduate thesis and dissertation submitted to universities in the United States of America and the United Kingdom, appears in three formats. The print version is called *Dissertation Abstracts International*; the online version is called *Dissertation Abstracts Online* while the compact disc format is called *Dissertation Abstracts Ondisc*. The full text of this publication is available online at http://www.proquest.com subject to certain protocols;
- The *African Studies Database* is a combination of abstracts and indexes of information on Africa. Available online (as BiblioLine on http://www.nisc.com) and on CD (as NISC DISCcover), this database is typically an index but also has abstracts of some of the items listed from three continents – Africa, Europe and North America.

3.8.2.3 Uses of Indexes and Abstracts

- For the selection and acquisition of library materials;
- They serve as aids for readers to evaluate the content of a document;
- They are used to facilitate retrieval of information by reducing the time users spend in locating information;
- They are used for compilation and production of a bibliography;
- They are tools used to identify articles in journals;
- They are also very important tools for researchers to know what materials have been published in their subject of interest.

Aina, L. O. (2004). *Library and Information Science Text for Africa*. Ibadan: Third world information Services Limited, pp239-247.

Bopp, R. E. & Smith, L.C. (Eds) (1995). *Reference and information services: an introduction*. Englewood, Colorado: Libraries Unlimited.

Borgman, C. L. (2000). *From Gutenberg to the Global Information Infrastructure*. Cambridge (MA): the MIT Press.

http://www.amazon.com/Blacks-Medical-Dictionary-42nd-Ed/dp/0713689021

Katz, W. A. (1997) *Introduction to reference works, Vol. 1: Basic Information Source* (7th ed.). New York: the McGraw-Hill Companies, Inc.,p436.

Lea, P. W. & Alan, D. (1996). *The Reference Sources Handbook* (4th ed.). London: LA Publishing.

Reitz, J. M. (2013). *Online Dictionary for Library and Information Science*. Retrieved from http://www.abc-clio.com/ODLIS/odlis_e.aspx on 28th August, 2013.

CHAPTER FOUR

ORGANISATION OF KNOWLEDGE

By

Christian Kofi, Wilfred Kobina Biney, William B. Baiden and Doris Osei-Agyemang

4.0 Introduction

This chapter deals with how information and for that matter knowledge is organized in the library for easy retrieval. Organising knowledge has to do with using tools in the structuring and dissemination of knowledge. Adequate knowledge on how materials are organised in the library is necessary for one to be able to search and retrieve the information one is looking for.

Library cataloguers have identified the following active verbs as fundamental needs by users in their quest for information namely: to find, to identify, to select and to obtain. Users find materials on a given topic and by a given author. In order to identify a record, users confirm that the record that is retrieved corresponds to the document or format that they are looking for. In selecting records, a user must have enough information to decide which of the multiple records best suits his needs either for language, format or both. Finally, a library user obtains the item after having enough information to find it on the shelf, order it or access it electronically.

Defining Information and Knowledge

Information is defined as facts or data that are collected, recorded, processed and stored in a retrievable form.

Knowledge refers to human inferences and interpretation derived from processed information. Therefore, knowledge is a blend of human experience, analysis, deduction and assimilation. As the quantity of knowledge has grown, so has it become increasingly important to develop tools for finding information and knowledge within a store or collection of information.

4.1 Tools for Organizing Information

The main tools used in organizing information are cataloguing and classification. They are two sides of the same coin.

Cataloguing & Classification

- Cataloguing and classification go together.
- They are two sides of the same coin.
- When you catalogue a book, you both catalogue (describe) and classify it to complete the process.
- Library catalogues exist in many forms and differ in detail according to type of library and users.
- Each bibliographic record in the catalogue represents an item.

4.1.1 Cataloguing

This is the process of recording information about an item according to accepted standards. Such information includes

author, title, edition, publisher, year, etc. When you catalogue a book, you both describe and determine its subject matter it to complete the process.

Reasons for Cataloguing Library Materials

1. To enable a library user find a book of which the author, title title or the subject is known.
2. To show what the library has by a given author, on a given subject and in a given kind of literature.
3. To assist library users in the choice of a book as to its edition or as to its character.

Issues Considered in Cataloguing and Classification

- Assign appropriate subject heading to an item, such as Economics, History, Zoology or Computer Science as the case may be.
- The Heading is taken from an authorized or standard list
- Large libraries use the Library of Congress Subject Headings (LCSH)
- Smaller ones use Sears List of Subject Headings.
- Use title page as the chief source of information
- Determine who is responsible for the intellectual or artistic content of the work.

Introduction to MARC

MARC is the acronym for Machine Readable Catalogue. "Machine-readable" means that one particular type of machine, a computer, can read and interpret the data in the cataloguing

record. MARC is a short and convenient term for assigning labels to each part of a catalogue record so that it can be handled by computer. MARC is primarily designed for meeting the needs of libraries; however this concept has been embraced by the wider information community as the convenient way of storing and exchanging bibliographic data. The MARC format was developed in the United States by the Library of Congress in 1965-66, but since then a number of formats have appeared with some common characteristics. Notable among them are the USMARC, UKMARC and other national MARC formats. Though all are meant for representing bibliographic items, some differences seem to exist in the content designation. The Library of Congress and the National Library harmonized the USMARC and CANMARC formats in a single edition in early 1999 under a new name MARC 21.

A MARC 21 record contains fields, each designated by three character tag, which can store information of two types: firstly, about the type of materials, the physical characteristics, specific bibliographic aspects, and so on, which is required to process records and secondly about the bibliographic data elements required for cataloguing such as the author, title, publisher, etc.

Cataloguing Record

Cataloguing record means a bibliographic record, or the information obtained after cataloguing a material. The record includes (not necessarily in this order): 1) a description of the item, 2) main entry and added entries, 3) subject headings, and 4)

the classification or call number. (MARC records often contain much additional information).

Description

Librarians follow the rules in Anglo-American Cataloguing Rules, 2nd ed., 2002 revision to compose the bibliographic description of a library item. This "description" includes the title, statement of responsibility, edition, material specific details, publication information, physical description, series, notes, and standard numbers.

Main Entry and Added Entries

AACR2 also contains rules for determining "access points" to the record (usually referred to as the "main entry" and "other added entries"), and the form these access points should take. Access points are the retrieval points in the library catalogue where patrons should be able to look up for the item. In other words, the rules in AACR2 are used to answer questions such as: For this book, should there be entries in the catalog for more than one author or more than one title? Should the title of the series be noted? How should the author's name be written? Is this a "title main entry" item (no author)?

International Standard Bibliographic Description (ISBD)

This was developed in Copenhagen in 1969 at the same time as MARC for bibliographic data; Separates the different parts of the bibliographic description of an item with; ISBD has been incorporated into most cataloging codes developed or modified

since 1969. The description is divided into eight (8) sections. The following constitute the sections:

- Title statement
- Edition statement
- General material designation (GMD)
- Imprint statement
- Physical description
- Series (collection) statement
- Notes
- Standard numbers (ex.: ISBN, ISSN)

Connections between ISBD, AACR & MARC

The fields and subfields in the MARC 21 formats correspond to the "areas" of ISBD; Chapters in AACR correspond to ISBD sections;

Changes to the rules (in general) require corresponding changes in the MARC formats, or at least review to consider the accommodation of the data resulting from the revised rule in the MARC data elements;

The three are used by the entire world; documentation has been translated in various languages.

Sections of ISBD and MARC 21 Fields:

- Title statement - Field 245
- Edition statement - Field 250

- General material designation - Field 340
- Imprint statement - Field 260
- Physical description - 3XX fields
- Series statements - 4XX fields
- Notes - 5XX fields
- Standard numbers (ISBN, etc.) - 02X fields

Resource Description Access (RDA) is the new cataloguing standard that will replace Anglo-American Cataloguing Rules, 2nd edition. RDA goes beyond earlier cataloguing codes in that it provides guidelines on cataloguing digital resources and a stronger emphasis on helping users find, identify, select, and obtain the information they want. RDA also supports clustering of bibliographic records to show relationships between works and their creators. This important new feature makes users more aware of a work's different editions, translations, or physical formats.

What is Metadata?

Metadata is structured information that describes, explains, locates, or otherwise makes it easier to retrieve, use, or manage an information resource. Metadata is often called data about data or information about information.

The term metadata is used differently in different communities. Some use it to refer to machine understandable information, while others use it only for records that describe electronic resources. In the library environment, metadata is commonly used for any

formal scheme of resource description, applying to any type of object, digital or non-digital. Traditional library cataloging is a form of metadata. MARC 21 and the rule sets used with it, such as AACR2, are metadata standards. Other metadata schemes have been developed to describe various types of textual and non-textual objects including published books, electronic documents, archival finding aids, art objects, educational and training materials, and scientific datasets.

There are three main types of metadata:

- **Descriptive Metadata** describes a resource for purposes such as discovery and identification. It can include elements such as title, abstract, author, and keywords.
- **Structural Metadata** indicates how compound objects are put together, for example, how pages are ordered to form chapters.
- **Administrative Metadata** provides information to help manage a resource, such as when and how it was created, file type and other technical information, and who can access it. There are several subsets of links to resources based on audience or topic.

Library Catalogue

A library catalogue is a list of documents in a library, with the entries representing the documents arranged in some systematic order. It provides data about items in the collection which the cataloguers represent. The library catalogue carries information on where the items can be found. Almost all library catalogues are

multiple access files. Multiple access files are entry points or ways to get to an item and retrieve that particular record.

The Primary Forms of Library Catalogues

These are Card catalogue, Book catalogue, Microform catalogue and Computer access catalogue/online catalogue.

- **Card Catalogues** consist of multiple drawers containing 3 X 5 cards arranged alphabetically by author, title and added entries. Added entries consist of cards for such things as more than one author, editors, compilers, translators, illustrators, arrangers of music, and series. Catalogues that are divided into sections are referred to as divided catalogues. Divided catalogues may consist of three sections (author, title and subject) or two sections (author/title). The cards in each section are filed word-by-word.

- **Book Catalogues** list the library's holdings in book form. It is difficult to keep these catalogues up- to-date and therefore, it is necessary for frequent supplements. With a book catalogue, it is rather easy to maintain several catalogues at the same time in different locations. The "updates" are kept in separate volumes until complete revisions are made.

- **Microform Catalogue** is a library's holdings listed on micro format. This usually requires a reading device like a microscope or microfilm reader.

Computer Access Catalogue/Online Catalogue is also referred to as Online Public Access Catalogue (OPAC). OPAC can be accessed on computer terminals within a library or on the Internet outside the Library.

Guide to Original Cataloguing

- Read the introduction
- Read the preface
- Read any review
- Go carefully through the content pages
- Examine the index
- Take a close look at the text and illustrations
- The author and/or the publisher may give some indication of the subject matter.
- Consult a fellow professional or scholar in the field.
- Express content as accurately as possible within the provisions in the scheme in use.

Example of Card Catalogue entry

• KF 9218. Sa4	1. Call mark (Class no. + Cutter no.)
• SAMAHA, Joel	2. Author(s)
• Criminal law/Joel Samaha	3. Title/Statement of Responsibility
• . - 4th ed.	4. Edition Statement
• Minneapolis: West Publishing Co. , 1993.	5. Publication: (place: publisher, year.)
• xviii, 577 p. : ill.	6. Physical Description (Collation, Pagination)
• 231712	7. Accession Number

• 0-314-00774-1	8. International Standard Book Number (ISBN)
• Criminal Law – United States – Cases	9. Subject Headings
• Includes bibliographical references and index	10.Notes and Bibliography

Access Points

An access point is a code, term, or the like through which an entry in a bibliographic record may be found. Essentially, access points are doorways or windows that can be used to trace a material in a catalogue or database.

Access to bibliographic records is provided through access points. Access points are specific pieces of information, such as call mark, author, title, subject (subject headings), series and ISBN. This standard information is common to all bibliographic records and is used to locate records or additional information.

Call Mark: is a number, letter, or combination of these used as symbols to indicate the specific location of a work in a library, especially the combination of the classification symbol and the designation for the author. It consists of the class number and the author or title cutter number.

Author: is the person(s) or corporate body responsible for the intellectual content of the item. It is the person or body that is responsible for creating the item. This means that editors,

corporations, government agencies, or meetings can also be considered authors. Some items have more than one author. If you know any author of the item you need, you would use that as your access point. Most catalogues require that the last name be entered first without a comma.

Title: is the identifying name of the intellectual work (book, play, journal, article, video, audio recording, etc.). Some items may have more than one title. You may use either as your title access point, as most libraries will provide title access using both titles. When you are searching by title, do not include insignificant words or definite and indefinite articles such as "the", "an", "a", at the beginning of the title.

Subject or subject heading is a word or phrase that describes the intellectual content of the item. Bibliographic records represent actual items and some items will be about more than one subject or more than one aspect of a subject. This means that records often have more than one subject heading. If you want an information about someone, something, or somewhere you would search by subject.

Series-A series statement appearing anywhere in the publication should be recorded. Eg. (Legon economics series; no. 42), (Macmillan student series).

International Standard Book Number (ISBN) – is a commercial identifier for books. The ISBN is 10 or 13 digits long

and it uniquely identifies internationally published books and book-like products. The number consists of 4 to 5 parts:

a. If a 13 digits ISBN, the number has a 978 or 979 prefix EAN (European Article Number) that ensures no two starts the same
b. Country or Language Code (0, 1 =English; 2 = French; and so on. Can be up to 5 digits)
c. Publisher (assigned by the agency)
d. Item Number
e. Check Digit (a single digit computed from other digits) if 10 it's replaced by "X"

4.1.2 Classification

The Meaning of Classification

Classification is a process of grouping. It involves putting together like entities and separating unlike entities. In ordinary classification, we deal with the arrangement of ideas and objects. But in library classification, we are concerned with documents and the aim is to arrange these in the most helpful order. According to Sayers (1975) it is "the arrangement of books on the shelves or descriptions of them, in the manner which is most useful to those who read". The emphasis is on usefulness, so that users can search for items without difficulty. Mann (1943) also notes that classification "is the arranging of things according to likeness and unlikeness. It is the sorting and grouping of things, but in addition, classification of books is a knowledge

classification with adjustments made necessary by the physical form of books.

Bibliographic Classification Systems

For the purpose of our study, one can take note of two major classification systems: Dewey Decimal Classification (DDC) and Library of Congress Classification. While the DDC is used by smaller libraries such as the public library and the school library, the LC is used by large libraries including academic and national libraries.

The Library of Congress Classification Scheme (LC)

The LC is basically a classification by discipline and consists of 21 main classes. The structure is as follows:

Outline of the Main Classes

- A - General Works
- B – Philosophy, Psychology, Religion
- C - Auxiliary Sciences of History
- D - History: General and Old World (EH)
- E-F - History: America (Western Hemisphere)
- G - Geography. Maps. Anthropology. Recreation
- H - Social Sciences
- J – Political Science
- K – Law (General)
- L - Education
- M - Music

- N – Fine Art
- P- Languages and Literature
- Q- Science
- R - Medicine
- S - Agriculture
- T - Technology
- U - Military Science
- V - Naval Science
- Z - Bibliography. Library Science

4.2 An Integrated Library System (ILS)

An integrated library system (ILS), which is also known as a library management system (LMS), is an enterprise resource planning system for a library, used to track items owned, orders made, bills paid, and patrons who have borrowed materials. An ILS usually comprises a relational database, software to interact with that database, and two graphical user interfaces (one for patrons, one for staff). Most Integrated Library Systems separate software functions into discrete programs called modules, each of them integrated with a unified interface. Examples of modules might include:

- acquisitions (ordering, receiving, and invoicing materials)
- cataloguing (classifying and indexing materials)
- circulation (lending materials to patrons and receiving them back)
- serials (tracking magazine and newspaper holdings)
- the OPAC (public interface for users)

Koha is an open source Integrated Library System (ILS), used world-wide by public, school and special libraries. The name comes from a Māori term for a gift or donation.

Koha is web-based ILS, with a Structured Query Language database (MySQL preferred) backed with cataloguing data stored in Machine Readable Catalogue (MARC) format and accessible via Z39.50. The user interface is very configurable and adaptable and has been translated into many languages. Koha has most of the features that are expected in an ILS, including:

- Simple, clear interface for librarians and members (patrons)
- Various Web 2.0 facilities like tagging, comment, Social sharing and RSS feeds
- Union catalog facility
- Customizable search
- Circulation and borrower management
- Full acquisitions system including budgets and pricing information (including supplier and currency conversion)
- Simple acquisitions system for the smaller library
- Ability to cope with any number of branches, patrons, patron categories, item categories, items, currencies and other data
- Serials system for magazines or newspapers
- Reporting
- Reading lists for members

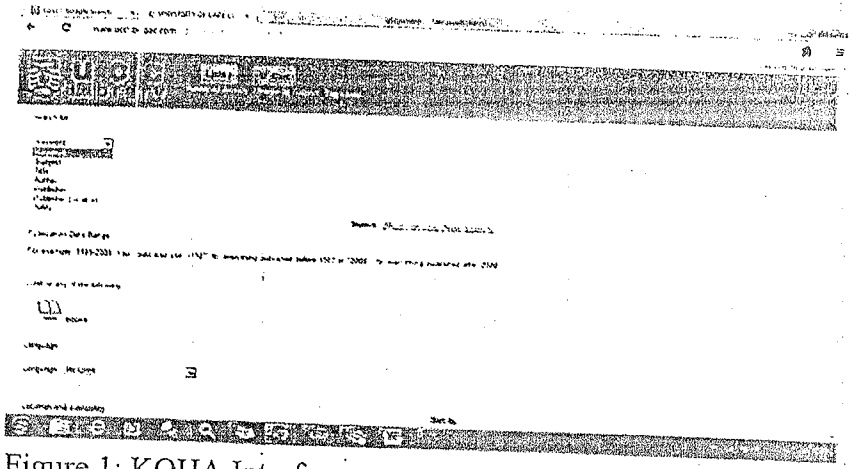


Figure 1: KOHA Interface

Figure 1 above shows Koha interface for patrons to enter their queries. Patrons have optional access points to select from namely Keywords, Title, Author, ISBN, among many others.

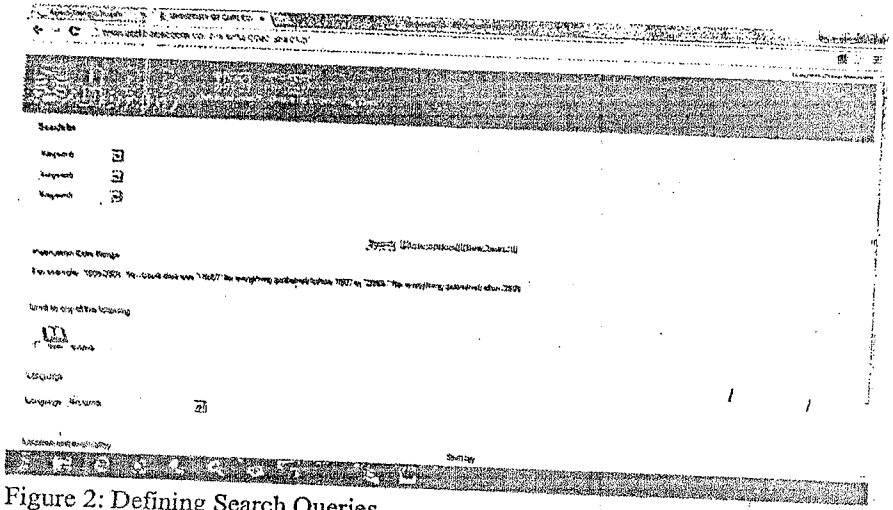


Figure 2: Defining Search Queries

The above interface shows where students can define their search queries using their selected key words. Key words must be keyword in context to enable easy retrieval (KWIC). A KWIC index is formed by sorting and aligning the words within a database to be searchable alphabetically except for the stop words.

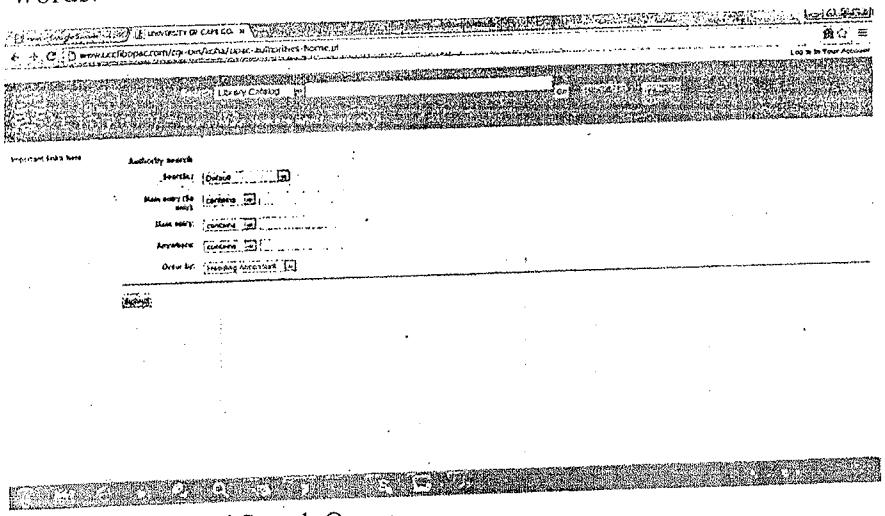


Figure 3: Advanced Search Query

This interface shows an advanced search query, patrons can combine the Boolean Operators of "AND", "NOT", "OR" to expand or limit their search results in order to achieve relevancy, specificity and precision.

References

Amanquah, S.N. (2000). *Introduction to cataloguing and classification.* Legon: University of Ghana.

Bolner, M. S. & Poirer, G. A. (1997). *The Research process.* Iowa: Kendall/Kunt Pub., Co.

Chowdhury, G. G. (2004). *Introduction to modern information retrieval* (2nd ed.) London: Facet Publishing.

Mann, M. (1943).*Introduction to cataloguing and classification of books.* (2nd. ed.). Chicago: American Library Association. p. 38.

Rowley, J. F. & Farrow, J. (2000). *Organizing knowledge: an introduction to managing access to information.* (3rd ed.) Cambridge: Cambridge University Press.

Sayers, W. C. B. (1975). *Manual of classification for librarians and bibliographers* (3rd ed.). London: Andre Deutsch.

Taylor, A. G. & Joudrey, D. N. (2008). *The organisation of information.*(3rd ed.).Westport: Libraries Unlimited.

Taylor, W. & William, H. K. R. (2010). *RDA: Resource Description and Access.* Ariadne, 63.

CHAPTER FIVE

ETHICAL ISSUES: COPYRIGHT, CITATION AND PLAGIARISM

By

Christian Kofi, Wilfred Kobina Biney, William B. Baiden and Doris Osei-Agyemang

5.0 INTRODUCTION

This chapter discusses ethical issues in writing such as copyright, citation and plagiarism and its consequences, and how information retrieved can be used legally.

5.1 COPYRIGHT LAW

The World Intellectual Property Organisation (WIPO) defines copyright as a legal term that describes the rights given to creators for their literary and artistic works. According to Harrod (1981), copyright refers to a procedure whereby the originator of a piece of intellectual property such as a book, article or piece of music receives due recompense for inventiveness or imagination expended. Copyright is the branch of law that holds the legal provisions on the rights of writers, artists and other creative persons to determine how their literary or artistic works may be used. For a creation to be regarded as a piece of work, it must have attained the high standard required of a work, meaning that

it should display originality or individuality as a result of the originators personal making. It is a legal term/concept that gives the creator of original work "exclusive rights" to control the use of his/her work for a limited time. These rights are:

- the right to make copies,
- the right to be credited for the work,
- the right to determine who may adapt the work to other derivative forms, such as translations, performance, financial benefit from the work, and other related rights.

Authors may transfer by contract those rights to publishers, or users, to authorize them to make copies, translations, performances, recordings...However, authors do not have monopoly to control all uses of their works: the law foresees some legal exceptions to the benefit of the public.

Main Issues in a Copyright Law

1. Gives monopoly to authors and creators in order to stimulate intellectual and artistic creativity.
2. Authors and creators may transfer their monopolistic rights to publishers for marketing purposes.
3. Users/Publics have rights in the form of fair use (academic purposes and knowledge improvement).

Items covered in the Copyright Law

In Ghana the following works are eligible for copyright protection: art, sound recordings, literature, musical works,

choreography, audio-visual works, and computer programs. In addition, any derivative works (any work based on an already copyrighted work) is also granted copyright protection. However, there is a specific criterion a work must meet in order to be protected. For example, the work must be original and must already be "fixed in definite medium of expression". Ideas or concepts are not entitled to copyright protection. Additionally, the creator must be a citizen of Ghana or the work must have been published in Ghana before anywhere else.

Three Copyright Acts of Ghana:

1. Copyright Act 85 of 1961
2. PNDC Law 110
3. Copyright Act 690 of 2005

The Library is mentioned under the third Section, Permitted Uses of Copyright Sub-Section 21 of the Copyright Law in Ghana.

1) A library or archive with activities that are not for gain may, without the authorization of the author of copyright, make a single copy of the work by reprographic reproduction.
2) A reprographic reproduction under subsection (1) may be made when the work reproduced is a published article, other short work or short extract of a work and where the purpose of the reproduction is to satisfy the request of an individual.
3) The library or archive shall under subsection (1) ascertain that the copy is to be used solely for the purpose of study, scholarship or private research.

4) The act of reproduction under subsection (1) a copy is made in order to preserve or replace a copy which has been lost, destroyed or rendered unusable in the permanent collection of similar library or archive if it is impossible to obtain the copy under reasonable conditions.
5) Where a library or, archive requires more than a single copy of a work by reprographic reproduction, the permission for this shall be obtained from the author, other owner of copyright or from an appropriate collective administration society authorized by the publisher.

Under Section 2 of the Bill, the following is stated:

1) The rights of the author are protected during the life of the author and seventy years after the death of the author.
2) Where a work is co-authored, the rights of the authors referred to are protected during the life of the last surviving author and seventy years after the death of that author.

5.2 CITATION STYLES

All fields of research agree on the need to document scholarly borrowings, but documentation conventions vary because of the different needs of scholarly disciplines. The most common citation styles are the American Psychological Association (APA) and Modern Languages Association (MLA). APA style for documentation is most commonly used to cite sources in social sciences while MLA is widely used in the humanities especially in writings on languages and literature. Another citation style, the Chicago (CS) is used in some social science publications and

most historical journals. It remains the basis for the Style Guide of the American Anthropological Association and for the Organisation of American Historians. It is used widely by academic and some trade publishers, as well as editors and authors who are required by those publishers to follow it. Chicago style offers writers a choice of several different formats. It invites the mixing of formats, provided that the result is clear and consistent. The Harvard citation style can vary in minor features such as punctuation, capitalisation, abbreviations, and the use of italics.

Referencing or citing your sources is an important part of academic writing. The purpose of teaching citation styles is to help students with their citation needs. It lets you acknowledge the ideas or words of others if you use them in your work and helps avoid plagiarism. Referencing also demonstrates that you have read relevant background literature and you can provide authority for statements you make in your assignments.

Each citation has given examples, and in some instances the citation style will have examples for the following: a single author book, book with two to six authors, edited books, journal articles, online journal articles, websites (multi page or portion) among others.

Types of Citation Styles

1. APA - American Psychological Association
2. MLA - Modern Language Association

3. Chicago / Turabian Style
4. Harvard (formerly known as AGPS)
5. AAA - American Anthropological Association
6. ACS - American Chemical Society
7. AMA - American Medical Association
8. APSA - American Political Science Association
9. ASA - American Sociological Association
10. American Antiquity
11. CSE (CBE) - Council of Biology Editors (now called the Council of Science Editors)
12. IEEE Documentation style (computer sciences)
13. Legal Citation
14. Statistics Canada
15. SAA - Society for American Archaeology
16. UBC Biology 140 Citations

In-Text Citations: Author/Authors

1. A work by two authors: Name both authors in the signal phrase or in the parentheses each time you cite the work. Use the word "and" between the authors' names within the text and use the ampersand (&) in the parentheses. For example, Research by Alemna and Badu (2005) showed..... or (Alemna & Badu, 2005)
2. A work by three to five authors: List all the authors in the signal phrase or in parentheses the first time you cite the source. For example, (Alemna, Badu, Kissiedu, Adjei & Akusah, 2006). In subsequent citations, only the first author's

name followed by "et al." in the signal phrase or in parentheses. For example, (Alemna et al. , 2006)

3. Six and more authors: Use the first author's name followed by et al. in the signal phrase or in parentheses. For example, Bannerman et al. (2001) argued.....or (Bannerman et al. , 2001)

APA Reference List: Author/Authors

The following rules for handling works by a single author or multiple authors apply to all APA style references in your reference list, regardless of the type of work (book, article, electronic resources, etc.)

APA Reference List: Books (Basic format)

1. Author (s)
2. (Year of publication)
3. Title of work: Capital letter to begin subtitle.
4. Location:
5. Publisher.
6. Page number.

APA Reference List: Articles in Periodicals

1. Author(s)
2. Year in brackets.
3. Title of article.
4. Title of Periodical,
5. Volume number (issue number),
6. Pages.

EXAMPLES

1. APA - American Psychological Association 5th edition

Book with a single author

Nkrumah, K. A. (2001). *The careful writer.* New York: Macmillan. p 25

Book with two to six authors

Boakye, J. O., Acheampong, F. O. & Erskine, G. (2005). *Introduction to educational psychology.* Accra: APH Publishing Corporation. p 73

Edited Book with no author

Baiden, W. & Osei-Agyeman, D. (Eds). (1997). *Consequences of growing up poor.* New York: Russell Sage Foundation. p 16

Edited book with author(s)

Fosu, V. S. (2000). *The unabridged journals.* (P. K. Yamoah, Ed.). Boston: CAB Publishers. p 136

Book with edition other than the first

Sakyi-Armah, K., Dadson, R. & Adjetey, F. (2010). *The battered child* (5th Ed.). Kumasi: KNUST Press. p 105

Chapter in an edited book

Bannerman, K. A. (2012). Family therapy. In G. A. Mensah (Ed.), *Handbook of family therapy* (pp. 742-775). New York: Springer.

Journal article (in print)

Afrifa, R. K., Frimpong, G., Amegashie, P.K. & Owusu, K. A. (2013). The prototype as a construct. *Journal of psychology*, 90, 57-68.

Journal articles from online databases

Kwarteng, A. M. (2008). The origin and spread of the calumet ceremony. *American Antiquity*, 46(4), 75-81. Retrieved October 3, 2010 from JSTOR database.

Websites

Multi page website

Ministry of Health, Task Force on Malaria Prevention and Eradication. *Facing the malaria canker heads on.* Retrieved May 23, 2003 from http://www.moh.gov.gh.

Portion of a Website

American Psychological Association. (1998). Managing traumatic stress: Tips for recovering from disaster and other traumatic events. Retrieved April 15, 2001, from American Psychological Association, Help Center Web Site: http://helping.apa.org/family/warning.html

2. MLA - Modern Language Association 6th edition

Book with a single author

Mahama, John D. My First Coup d'etat: Memories From the Lost Decades of Africa. London: Bloomsbury, 2012.

Books with two or three authors

Asamoah, Franklin A., Martin K. Quartey and Obed O. Ankamah. The Two-Year College. Cape Coast: UCC Press, 2015.

Books with four or more authors

Afful, Bernard K. et al. Teaching Shakespeare. Legon, Accra: UG Press, 2000.

Edited book

Gogovi, Ellis and James Takyi-Mensah, eds. Healing from within: The use of Hypnosis in Women's Health Care. Washington D.C: Jossey-Bass, 2002.

Journal article (in print)

Obeng, Evans K. "Hearers and Speech Acts". Language, 58 (1982): 332-373.

Journal articles from online database

Lartey, Georgette. "The Common Reader's Shakespeare". Early Modern Library Studies 3.3 (1998): 12 pars. 15 Aug. 2003.

Websites

Multi page website

Early Modern English Dictionaries Database (EMEDD). Ed. Isaac Lamptey. Apr. 2000. University of Cape Coast. 20 Sept. 2003.

Portion of a Website

"BC and Canada Immigrant Landings by Class - 1984-2001". Migration and Immigration Feb. 2001. BC Stats. 15 Aug. 2003.

3. Chicago Style

Book with Single Author

Footnote:

1. Ann Oakley, *Gender on Planet Earth* (New York: New Press, 2003), 77.

Bibliography:

Oakley, Ann. *Gender on Planet Earth.* New York: New Press, 2003.

Book with Two or Three Authors

Footnote:

Marilyn J. Boxer and Jean H. Quataert, *Connecting Spheres: European Women in a Globalizing World, 1500 to the Present,* 2nd ed. (New York: Oxford University Press, 2000), 133.

Bibliography:

Boxer, Marilyn J. and Jean H. Quataert. *Connecting Spheres: European Women in a Globalizing World, 1500 to the Present.* 2nd ed. New York: Oxford University Press, 2000.

Book with More than Three Authors

Footnote:

M. Clayden et al., *Making it Work: A Handbook for Reading, Writing, Language and Media* (Toronto: Irwin Publishing, 2003), 65.

Bibliography:

Clayden, Marie, Dianne Fenner, Christine McAdam, and Christine Strauss.*Makingit Work: a Handbook for Reading, Writing, Language and Media.* Toronto: Irwin Publishing, 2003.

Edited Book

Footnote:

Lorna Crozier, "What Stays in the Family," in *Dropped Threads: What We Aren't Told,* ed. Carol Shields and Marjorie Anderson, 11-18 (Toronto: Vintage Canada, 2001).

Bibliography:

Crozier, Lorna. "What Stays in the Family." In *Dropped Threads: What We Aren't Told,* edited by Carol Shields and Marjorie Anderson, 11-18. Toronto: Vintage Canada, 2001.

Journal Article (hard copy)

Footnote:

Benjamin Berger, "The Limits of Belief: Freedom of Religion, Secularism, and the Liberal State," *Canadian Journal of Law and Society* 17, no. 1 (2002): 61.

Bibliography:

Berger, Benjamin. "The Limits of Belief: Freedom of Religion, Secularism, and the Liberal State." *Canadian Journal of Law and Society* 17, no 1 (2002): 39-68.

Journal Article (electronic version)

Footnote:

Gerald P. Hodgkinson, "The Interface of Cognitive and Industrial Work and Organisational Psychology," *Journal of Occupational and Organisational Psychology* 76, no. 1 (2003), http://www.epnet.com.

Bibliography:

Hodgkinson, Gerald P. "The Interface of Cognitive and Industrial Work and Organisational Psychology." *Journal of Occupational and Organisational Psychology* 76, no. 1 (2003): 1-25. http://www.epnet.com.

Websites

Footnote:

George F. MacDonald, "The Haida: Children of Eagle and Raven," Canadian Museum of Civilization Corporation, http://www.civilization.ca/aborig/haida/haindexe.html.

Bibliography:

MacDonald, George F. "The Haida: Children of Eagle and Raven." Canadian Museum of Civilization Corporation. http://www.civilization.ca/ aborig/haida/haindexe.html.

4. Harvard Citation Style

Writing References

Kennedy, R.R., 2009. The power of in-class debates. *Active learning in higher education,* 10 (3), 225-236.

Aldrick, P., 2011. Christmas cheer for high street amid recession warning. *The Daily Telegraph*, 21 December 2011, 3.

Edited work

Briassoulis, H., 2004. Crete: endowed by nature, privileged by geography, threatened by tourism? In: Bramwell, B., ed. *Coastal mass tourism: diversification and sustainable development in Southern Europe.* Clevedon: Channel View, 48-67.

If there is more than one contributing author who wrote the chapter, you must list all authors in the reference list at the end of your work e.g. Smith, A., Jones, T. and Bloggs, J., 2010...

One Author

Woods, P., 1999. *Successful writing for qualitative researchers.* London: Routledge.

In this example, you need to look on the back of the title page to find the date.

If the book is edited, for example if Peter Woods had edited this book, then the reference would read Woods, P., ed. 1999...

The use of 'ed.' or 'eds.' (for more than one) is used to represent the word 'editor'.

Two Authors

Brewerton, P. and Millward, L., 2001. *Organisational research methods: a guide for students and researchers.* London: Sage.

Three or More Authors

McNiff, J., Lomax, P. and Whitehead, J., 1996. *You and your action research project.* London: Routledge.

Print Conference Paper

Granger, S., 1994.The hacker ethic. In: Kizza, J.M., ed. *Ethics in the Computer Age,* 11-13 November 1994 Tennessee. New York: ACM Press, 7-9.

Print Patent

Mavin, G. and Stevenson, T., (Portola Packaging Limited), 2007. *Container closure in combination with a ring pull insert.* UK patent application 2428669 A. 7 February 2007.

Print Reports

Kinnersly, R., 2013. *A generic guide for small islands on the implications of signing up to the Convention on Biological Diversity.* Peterborough: Joint Nature Conservation Committee. Report 489.

Print Theses

Young, V. E., 1997. *Special collections in the year 2015: a Delphi study.* Thesis (PhD). University of Alabama.

Article in an Online Journal

Kennedy, R.R., 2009. The power of in-class debates. *Active learning in higher education* [online], 10 (3), 225-236.

If there are multiple authors, you must list all authors in the reference list at the end of your work e.g. Kennedy, R.R. and Smith, T., 2010...

5.3 PLAGIARISM

Plagiarism means claiming or implying original authorship of (or incorporating material from) someone else's written or creative work, in whole or in part, into one's own. It deals with e.g. pictures, text, graphs or diagrams. Plagiarism occurs by direct

copying or by repetition without stating the source. As long as the source can be accounted for satisfactorily, a summary or a representation of another writer's piece of work is not considered as plagiarism. It also means presenting the words, phrases, ideas or work of another, including certain facts and statistics, as if they were your own.

As a student, you are expected to create original work to fulfill the requirements of the courses you take. Fair evaluation can occur only when submitted work reflects each student's efforts and aptitude. As a scholar, you are expected to cite the sources you use to support your arguments. Acknowledging the works and ideas of others demonstrates respect for the contributions of other authors and confers value to your own original work.

To avoid plagiarizing, you must clearly acknowledge the source of any borrowed language or ideas that you present in your own work. Quotation marks, followed by documentation, should be used to indicate the exact words of others. A signal phrase identifying a source and/or parenthetical citation or a superscript number should denote the summarized or paraphrased ideas of others, depending on the particular citation style the paper follows.

Plagiarism is therefore academic fraud or literary theft and puts one's academic integrity into disrepute. It is the practice of taking someone else's work, idea, invention, etc. and passing it off as one's own.

A student who is considered guilty of plagiarism will receive a caution and risks being suspended from the institution.

Examples of Plagiarism

Copying a text from the Internet, from a book or an article, without citing the source and handing it in as your own;

- Rewriting a text in your own words without stating your source;
- Paying for an essay and presenting it as your own;
- Handing in another student's work under your own name;
- Copying statistics, maps, tables or charts without stating your source;
- State someone else's artistic work as your own.

Strategies for Success

To avoid plagiarizing accidentally:

- Keep accurate notes when doing research
- Educate yourself on plagiarism policy
- Master the procedure for citing sources; and
- Learn to paraphrase without borrowing the language or structure used in an original source.

To avoid the temptation to plagiarize intentionally:

- Visit your instructor during office hours and ask questions in class to clarify course requirements or instructions for assignments;

- Consult the web-based resources in the webliography that follows. These sources will help you through the stages of writing a research paper, from planning to polishing;
- Consider the consequences of plagiarism before you try passing off the work of another as your own.

Consequences of Plagiarism / Penalties

The consequences or penalties for cases of plagiarism range from disciplinary sanctions such as probation, suspension, and dismissal to educational interventions such as attending a workshop or writing a paper. In addition, any work that is the product of plagiarism is generally assigned a grade of zero by the instructor.

If you are caught plagiarizing, it is possible that not only will you be removed from your course and institution, but you may never be able to study for a degree again. The problem is that students today are faced not only with increasing pressure and a consequent temptation to plagiarize, but also with the potential to plagiarize being ever more available via the Internet amongst other things. Since plagiarism is on the increase, you should not fall into the trap of thinking that because some people are tempted to plagiarize this somehow reduces its seriousness and therefore, the likelihood of you facing plagiarism consequences when caught.

The consequences of plagiarism could seriously endanger your academic career because your record will follow you, and should

you be compelled to leave your course of study, then you will find it extremely difficult to obtain a place elsewhere because no college or university will want a student who has proved himself to be dishonest and unworthy. These may seem very harsh words but it is no exaggeration to say that for a moment of foolishness you may well be sacrificing your academic career.

Should you embark on a longer piece of work, such as a dissertation or thesis, and plagiarize, the work will probably be destroyed. In fact, in these cases, your work will be scrutinized even more carefully for any instance of plagiarism and the consequences of any plagiarism discovered will be even more severe because originality is the basis of this type of high-level research and you will be required to produce work of publication standard.

References

Asamoah-Hassan, H. R. (2006). Copyright in Ghana. Retrieved from dspace.knust.edu.gh/jspui/bitstream/ on 20th February, 2014.

Purdue OWL APA formatting style guide. Retrieved from https://owl.english.purdue.edu/owl/resource/560/01/ on 25th February 2014.

Smith, A. N. & Donald B. M. (1987). *Information resources management.* Cincinnati: South-Western Publishing.

What is MLA Style? - Modern Language Association. Retrieved from http://www.mla.org/style on 25th February, 2014.

World Intellectual Property Organisation (WIPO) (1967) Retrieved from http://www.wipo.int/about-ip/en/copyright on 20th July, 2013.

CHAPTER 6

USING THE INTERNET TO ACCESS INFORMATION

By

Kwame Akonnoh Boohene, Paulina Nana Yaa Kwafoa, Wilfred Kobina Biney and Paul Nunekpeku

6.0 Introduction and Definition of Terms

This chapter deals with what the Internet is, the history, accessing the Internet, and uses and misuses of the Internet.

Internet

The Internet, also called the Net, is simply defined as a global network of computer networks. It is a large computer network that has connections to a vast number of computers all around the world that links millions of businesses, government agencies, educational institutions, and individuals. It is indeed made up of hundreds of thousands of connected networks around the globe. The Internet is not centrally controlled, and that means there is no regulatory body - no single person, company, institution, or government agency that controls the entire Internet. Thus, owners of various parts of the networks on the Internet maintain them.

Computer Networks

A computer network basically refers to a group of computers connected to each other, so they can share information and electronic equipment/resource. Computer networks actually allow

computers to share resources such as hardware (printer, webcam, scanner, etc), software, data and information. Computer network connections are either done by cables, wireless, telephone, satellite, etc. A computer network could be made up of two to over a million computers. In many networks, one or more computers act as a server or host. A server controls access to the resources on a network; and also provides services and connections to other computers on the network. The other computers on the network are called clients or workstations. The network could be in one locality or a limited geographic area; in that case we will have a Local Area Network (LAN) eg. interconnected computers and devices in small areas such as a home, a school, a computer laboratory, one office, one building, or a group of buildings close together. If the network spans over a wide geographic area such as a region/state or country, then we have a Wide Area Network (WAN). WANs are used to connect local area networks together, so that users and computers in one location can communicate with users and computers in other locations (eg. bank networks). There are other networks such as Campus Area Network (CAN) and Metropolitan Area Network (MAN). CAN is made up of interconnections of local area networks within a large campus or the LANs of different offices on a university campus linked together. In this case, the networking equipments (switches, routers) and transmission media (fiber optics, copper cable, Cat5 or Cat6 cabling, etc.) are entirely owned by the authorities like a University or an enterprise (for example Googleplex and Microsoft's campus). MAN is a large computer network that usually spans over a city, suburb or

metropolitan area and interconnects a number of separated local area networks within the same geographical area or metropolis using a high-capacity backbone technology, such as fiber-optical links, and provides up-link services to wide area networks and the Internet.

Protocols

Internet protocols are the set of rules that help the different computer networks on the Internet to communicate without conflict. They are the rules and encoding specifications for sending data. The protocols also determine whether the network uses a peer-to-peer or client/server architecture. The standard Internet communication protocol is called Transmission Control Protocol (TCP) and Internet Protocol (IP). The Internet protocol suite is often referred to as TCP/IP. The first major version of IP, Internet Protocol Version 4 (IPv4), is the dominant protocol of the Internet. Its successor is Internet Protocol Version 6 (IPv6).

6.1 History of the Internet

In the 1960s, researchers began linking computers to each other through telephone hook-ups, using funds from an agency known as Advanced Research Projects Agency (ARPA) of the U.S. Defense Department. The Internet which is a modern phenomenon became popular from the latter part of 1960s. Shelly and Vermaat (2011) state that the goal of the initial networking project called ARPANET was (1) to build a network, which allowed scientists at different locations to share information and work together on military and scientific projects. This was

basically to support military research and (2) to build a network which could function even if part of the network were disabled or destroyed by a disaster. It was to provide a measure of security against partial cable outages. The ARPANET could only be used to communicate.

The original ARPANET consisted of four main computers, one each located at the University of California at Los Angeles, the University of California at Santa Barbara, the Stanford Research Institute, and the University of Utah. Each of these computers served as a host on the network (Shelly and Vermaat, 2011). Numerous modification to this network resulted in the development of the World Wide Web (www) in 1989. The World Wide Web made the Internet interactive and also gave it a multimedia nature (that is made it possible to host text, sound and graphics). In 1990, the Hypertext Transfer Protocol (http) was created to enable the international networks to be used for web browsing. In 1995, Ghana was among the first countries in Sub-Saharan Africa to have Internet access through Network Computer Systems (NCS).

6.2 Accessing the Internet

Accessing the Internet basically involves the process of getting individual computers, mobile devices, and various computer networks connected to the Internet through the services of Internet Service Providers (ISPs). ISPs offer Internet access to the public through various technologies that offer a wide range of data

signaling rates (speeds). Broadly, connecting to the Internet could be grouped into two; Dial-up Access and Broadband Access.

Dial-up Connection to the Internet

This refers to the connection of a personal computer to the Internet using dial-up modems. Dial-up access uses a modem and an analog phone call placed over the Public Switched Telephone Network (PSTN) to connect to a pool of modems operated by an Internet Service Provider (ISP). The modem takes the digital information that your computer generates and turns it into analog information that can travel along the telephone network to other computers in remote locations. The Computers on the dial-up network must have dial-up modems to convert digital signals to analog and vice versa.

A dial-up connection operates on a single channel which monopolizes the phone line. It is the slowest method of accessing the Internet. Dial-up connections do not exceed a speed of 56 kilo bits per seconds (kbps), as they are primarily made using modems that operate at a maximum data rate of 56 kbps downstream (towards the end user) and 34 or 48 kbps upstream (toward the global Internet).

Broadband Connection to the Internet

This refers collectively to faster or high speed connections to the Internet. These may be a more costly options compared to the dial-up. The broadband when connected is always on and does not

require dialing up to the ISP to connect. Broadband provides improved access to the following Internet services such as:

- Faster world wide web browsing;
- Faster downloading of documents such as photographs, videos, music and other large files;
- Internet Telephony, radio, television, and video conferencing;
- Virtual Private Networks(VPN) and remote system administration;
- Online gaming.

6.2.1 Types of Broadband Connections

Digital Subscriber Line (DSL) Modem Connection

Digital Subscriber Line (DSL) is a family of technologies that provide connection to the Internet through a local telephone network. DSL uses regular phone lines, a DSL modem, and special technology to transmit data in mega bits per second (mbps). Unlike the dial-up, DSL can operate using a single phone line without preventing normal use of the telephone line for voice phone calls. DSL uses the high frequencies, while the low frequencies of the line are left free for regular telephone communication. Asymmetric Digital Subscriber Line (ADSL) is the most commonly installed variety of DSL. The data rates or speed to the consumer (downstream) ranges from 256 kbps to 20 mbps. The data rate in the direction to the service provider (upstream) is lower than the data rates to the customer, hence the designation "asymmetric." With a Symmetric Digital Subscriber

Line (SDSL), data rate in the direction to the service provider and the consumer are equal. Very-High-bit-rate Digital Subscriber Line (VDSL or VHDSL, ITU G.993.1) is a DSL standard that provides data rates up to 52 mbps to the consumer and 16 mbps to the service provider over copper wires and up to 85 mbps to the consumer and service provider on coaxial cable. VDSL is capable of supporting applications such as high-definition television as well as telephone services [Voice Over Internet Protocol (VOIP)] and general Internet access over a single physical connection. VDSL2 (ITU-T G.993.2) is a second-generation version and an enhancement of VDSL. It is able to provide data rates exceeding 100 mbps simultaneously in both the upstream and downstream directions. However, the maximum data rate is achieved at a range of about 300 meters.

Satellite Modem Connection

Satellites can provide fixed, portable, and mobile Internet access. It is among the most expensive forms of broadband Internet access, but may be the only choice available in some remote areas. Data rates range from 2 kbps to 1 gbps downstream and from 2 kbps to 10 mbps upstream. Satellite communication typically requires a clear line of sight. It will not work well through trees and other vegetation and it is adversely affected by moisture, rain, and snow. It may require a fairly large and carefully aimed directional antenna.

Cable Modem Connection

Cable Internet or cable modem access provides Internet access via hybrid fiber coaxial wiring originally developed to carry television signals. Either fiber-optic or coaxial copper cable may connect a node to a customer's location at a connection known as a cable drop. In a cable modem termination system, all nodes for cable subscribers in a neighborhood connect to a cable company's central office, known as the "head end." The cable company then connects to the Internet using a variety of means – usually fiber optic cable or digital satellite and microwave transmissions. Like DSL, broadband cable provides a continuous connection with an ISP.

Wireless

Wireless is another type of broadband that provides both fixed and mobile Internet access. Wireless Fidelity (Wi-Fi) can be used for a wireless local area network that uses IEEE 802.11 standards. Wi-Fi is used to connect laptops and smart phones to the Internet. It can be used to create campus-wide and city-wide wireless networks. Wi-Fi networks are built using one or more wireless routers called access points. The Wi-Fi network is connected to the larger Internet using DSL, cable modem, and other Internet access technologies. Data rates range from 6 to 600 mbps. Wi-Fi service range is fairly short, typically 20 to 250 meters or from 65 to 820 feet. Both data rate and range are quite variable depending on the Wi-Fi protocol, location, frequency, building construction, and interference from other devices. Using directional antennas, and with careful engineering, Wi-Fi can be extended to operate

over distances of up to several kilometers like wireless ISP. In addition, compared to hard-wired connectivity, there are security risks (unless robust security protocols are enabled). The network can be less stable, due to interference from other wireless devices and networks, weather and line-of-sight problems.

Mobile Broadband

Mobile broadband is a term for wireless Internet access delivered through mobile phone towers to computers, mobile phones, and other digital devices using portable modems. Some mobile services allow more than one device to be connected to the Internet using a single cellular connection through a process called tethering. The modem may be built into laptop computers, tablets, mobile phones, and other devices, added to some devices using PC cards, USB modems, and USB sticks or dongles. Alternatively, separate wireless modems can be used.

Bandwidth

Bandwidth determines the speed and amount of data or information that is transferred from one place to the other on the Internet. It is measured in bits per second (bps). Data transmission speeds could either be measured in kilo bits per second (kbps), megabits per second (mbps), or gigabits per second (gbps). For example, a network may connect at 512 Kbps, which means it can nominally send and receive 512,000 bits of information per second. The bigger the bandwidth size, the greater the volume of information that could be transmitted at a time, and the faster it

will be to move information from one place to the other on the Internet.

Devices Available for Accessing Information on Internet

To browse the Web, one needs a computer or mobile device that is connected to the Internet and has a Web browser. There are several personal computers (PCs) or devices that can be used to access information on the Internet. A personal computer is a computer that can perform all of its input, processing, output, and storage activities by itself. Some of these devices are as follows:

- Desktop/Tower Computer is designed so that the system unit, input devices, output devices, and any other devices fit entirely on or under a desk or table. In some models, the monitor sits on top of the system unit, which is placed on the desk. The most popular style of system unit is the tower, which can sit on the floor vertically.
- Laptop or Notebook Computer is a portable, personal computer often designed to fit on your lap. Notebook computers are thin and lightweight, yet can be as powerful as the average desktop computer.
- Netbook is a type of notebook computer which is smaller, lighter, and often not as powerful as a traditional notebook computer.
- Tablet PC or tablet computer is a special type of notebook computer you can interact with by touching the screen with your finger or a digital pen. One type of Tablet PC, called a convertible tablet, has an attached keyboard. Another type,

which does not include a keyboard, is called a slate tablet and provides other means for typing. Tablet PCs are useful, especially, for taking notes in locations where the standard laptop/notebook computer is not practical.

- Smart Phone is an Internet-enabled phone that also provides personal information management functions such as a calendar, an appointment book, an address book, a calculator, and a notepad. In addition to basic phone capabilities, a smart phone allows you to send and receive email messages and access the Web.
- E-book Reader (short for electronic book reader), or e-reader, is a handheld device that is used primarily for reading e-books. An e-book, or digital book, is an electronic version of a printed book, readable on computers and other digital devices. Most e-book readers have a touch screen and are Internet-enabled.
- iPad is a device that runs Apple's iOS. The user interface is built around the device's multi-touch screen, including a virtual keyboard. The iPad has in-built Wi-Fi and, on some models, cellular (chip) connectivity. It can shoot video, take photos, play music, and perform Internet functions such as web-browsing and emailing. Other functions of an ipad are games, reference works like dictionary, GPS navigation, and social networking like Facebook and Yookos.

6.2.2 Uses of the Internet

There are several uses or advantages of the Internet. It is widely used by many to meet different needs. The popularity of the

Internet has contributed, in most part, to the development of society. People are nowadays becoming more reliant on the Internet for their day to day work to save time and cost. When we summarize the usage of the Internet there are some common uses too. These are:

Communication

The foremost target of the Internet has always been for communication purposes. It has excelled beyond the expectations and still there are innovations going on to make it faster and more reliable. Whether at home or at work, some of the means by which we can use the Internet to communicate are through the Electronic mail (Email), mailing list, chatting, blogging, video conferencing, voice calls, social networks such as LinkedIn, Twitter, Facebook, Yookos, etc. Communicating on the Internet is currently the most convenient, cost effective and fastest method to interact with friends, family, and colleagues. For example, sending an email costs lower than posting a letter in the mail, especially for people who communicate internationally. The recipient also receives the email just in a matter of seconds the moment the mail is sent.

In fact, the Internet has become a great source to meet new people, and build new friendships and relationships. Discussion forums available on the Internet are useful to users to find individuals with the same interest in a particular topic. Interviews can be conducted online through video conferencing giving the same effects as real life interviews with no issues of being late

due to traffic situations on the road when traveling to the venue of the interview.

Information Dissemination

Information is probably the biggest advantage the Internet is offering. The Internet is a virtual treasure trove of information. There is a huge amount of information available on the Internet for just about every subject known to man, ranging from governance, law and services, trade fairs and conferences, market information, new ideas and technical support, the list is endless. An example of dissemination of information is in the fields of advertising, public announcements and speeches.

Education and Research

The Internet has become the largest and most useful source of information. It provides avenues for formal and informal teaching and learning. In this modern era, there are lots of online distance education courses ranging from certificate courses to doctorate degree courses, which are offered through the use of the Internet for a more structured learning experience. Students use the Internet to research for the purpose of gathering information. Teachers or lecturers give assignments that require research on the Internet as well as conduct examinations online. Others use the Internet to research for information virtually in every field of knowledge, and also to train employees to increase their productivity in their work.

Entertainment

Entertainment is another popular reason why many people prefer to surf the Internet. If you ask some people what their hobbies are today, some may reply as to 'browsing the Internet'. In fact, the Internet is a good place for fun. There are plenty to keep you occupied in your spare time. Many use the Internet to download free versions of software and cracks, listen to music, download movies, read news, play online games, visit chat rooms to join a community group or just surf the Web.

Online Services

An online service refers to any information and services provided over the Internet. These services do not only allow subscribers to communicate with each other, but also provide unlimited access to variety of information. Online services can be simple to complex. A simple online service may help subscribers have access to needed data through a search engine, while a complex one might be an online loan application from a bank. Online services may be free or paid for. Many services are now provided on the Internet, some examples are online banking, job seeking, online advertisement, purchasing tickets for your favourite movies, guidance services on array of topics, making hotel reservations, booking flights, up-to-date stock quotes, and articles from online journals.

E-Commerce

Electronic commerce, commonly referred as e-commerce, is one of the most effective and useful ways of conducting business. E-

commerce is the concept used for any type of commercial maneuvering, or business deals that involves the transfer of information across the globe via Internet. It is a type of industry where the buying and selling of products or services is conducted over electronic systems employing technologies such as email mobile devices, social media, and telephones as well. E-commerce is generally considered to be the sales aspect of e-business. It has become a phenomenon associated with any kind of shopping, almost anything. It has got a real amazing and wide range of products from household needs, technology to entertainment. The major different types of e-commerce are: business-to-business (B2B); business-to-consumer (B2C); business-to-government (B2G); consumer-to-consumer (C2C); and mobile commerce (m-commerce).

6.2.3 Misuses of the Internet

As much as there are enormous advantages in using the Internet, there are also some disadvantages associated with its usage. Some of these are:

Pornography

Pornography is perhaps one of the threats to society and also to healthy mental life. Internet pornography is pornography that is accessible over the Internet, primarily via websites, peer-to-peer file sharing, or Usenet newsgroups. There are thousands of pornographic sites on the Internet that can be easily found and accessed without restrictions. The Internet enables people to view pornography anonymously in the comfort and privacy of their

homes. It also allows access to pornography by people whose access is otherwise restricted for legal or social reasons, children for example.

Identity Theft/Fraud

If you use the Internet, you may be facing grave danger as your personal information such as name, address, credit card or VISA card number, email password and so on can be accessed by criminals to cause you problems. Some people use Internet services or softwares with Internet access to defraud victims or to otherwise take advantage of them by stealing personal information, which can even lead to identity theft. Identity theft is a form of stealing someone's identity in which one pretends to be someone else by assuming that person's identity. Usually, the impersonator accesses resources or obtains credit and other benefits in that person's name. This can make the victim suffer adverse consequences if he/she is held accountable for the perpetrator's actions. A very common form of Internet fraud is the distribution of rogue security software (fraud tool). Internet services can be used to present fraudulent solicitations to prospective victims, to conduct fraudulent transactions, or to transmit the proceeds of fraud to financial institutions or to others connected with the scheme.

Spamming

Spamming refers to sending of unsolicited and unwanted bulk messages. It is usually common with commercial emails. Individual recipients just receive lots of junk mails that they can

delete. However, spam also includes malicious emails that can contain viruses along with fraudulent emails (phishing) that pretend to be as an email from a legitimate source. Phishing is the act of stealing login information, often by sending emails that appear to come from a legitimate source such as a bank or websites like eBay. Such emails originate from criminals and are designed to trick you out of your login information. Softwares such as browser add-ons and plugins for your email client can help filter spam and phishing emails out of your inbox. For example, emails with subject headers such as "your business development pack" or "you have won a lottery" could all be spam if they are unsolicited.

Computer Virus Threat

A computer virus is a program or code that is loaded onto your computer without your knowledge, which disrupts the normal functioning of your computer systems. All computer viruses are man-made. Viruses often perform some type of harmful activity on infected computers, such as stealing hard disk space, use all available memory and bring the system to a halt, accessing private information, corrupting data, displaying political or humorous messages on the user's screen, or locking their keystrokes. Computer viruses can be prevented by installing antivirus on your computer and regularly updating.

References

Donahue, G. A. (2007). *Network warrior*. Farnham: O'Reilly.

E-commerce and e-business. Retrieved from http://en.wikibooks.org/wiki/E-Commerce_and_EBusiness/Concepts_and_ Definitions on 20th June, 2013.

Internet access. Retrieved from http://en.wikipedia.org/wiki/Internet_access on 14th June, 2013.

Junk email and spam. Retrieved from http://www.ehow.com /internet/email/junk-email-spam/#ixzz2ioh5yXCH on 20th June, 2013.

Long, C. (2001). *IP network design*. Osborne: McGraw-Hill.

Online service. Retrieved from http://www.techopedia.com/ definition/3248/online-service on 20th June, 2013.

Online service. Retrieved from http://www.webopedia.com/ TERM/O/online_service.html on 20th June, 2013.

Parsons, J. J. & Oja, D. (2002). *New perspectives on computer concepts* (5th ed.). Boston: Course Technology.

Shelly, G. B. & Vermaat, M. E. (2011). *Discovering computers 2011: living in a digital world, complete*. Boston: Cengage Learning.

Shelly, G. B. & Vermaat, M. E. (2012). *Discovering computers – Fundamentals: your interactive guide to the digital world*. Boston: Cengage Learning.

The difference between a computer cirus, worm and trojan horse. Retrieved from http://www.webopedia.com/DidYou Know/Internet/2004/virus.asp on 20th June, 2013.

What does dissemination of information mean? Retrieved from http://references-definitions.blurtit.com/35904/what-does-dissemination-of-information-mean on 20th June, 2013.

Williams, B.K. & Sawyer, S. C. (2011). *Using information technology: a practical introduction to computers and communications* (9th ed.). New York: McGraw-Hill.

CHAPTER 7

SEARCHING TOOLS ON THE INTERNET

By

Paulina Nana Yaa Kwafoa, Kwame Akonnoh Boohene, Wilfred Kobina Biney and Paul Nunekpeku

7.0 Introduction and Definition of Terms

This chapter deals with the various steps for performing effective searches on the Internet. The terminologies such as browsers, search engines and databases are explained. It also includes the evaluation of information obtained on the Internet and information and Web2 technologies.

Browsers

A web browser (commonly referred to as a browser) is a software or a program that is used to navigate or access the Internet. Although browsers are primarily intended to be used on the World Wide Web, they can also be used to access information provided by web servers in private networks or files in file systems. Browsers can also be used for communication and retrieval of information. It is however important to know that without a browser one cannot be able to access the Internet.

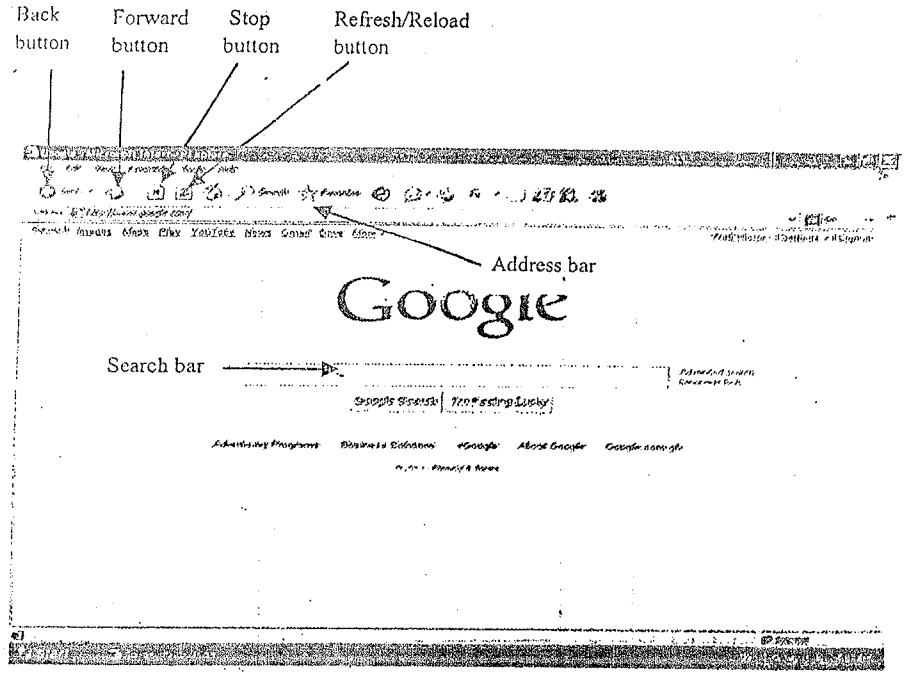
There are two types of browsers namely, text-based and graphic-based browsers. Text-based browsers provide access to only text within web pages, and ignore graphic content. They usually

render web pages faster than web browsers that include graphics. Text-based browsers could be very useful to users with visual impairment, since it takes less time for the computer to process and present especially with speech synthesis or text-to-speech software which reads content to users. Examples of text-based browsers are Lynx and Mosiac. Graphic-based browsers on the other hand render web pages with text, graphic, video or audio content. They support the multimedia nature of the Internet. Some examples are Mozilla Firefox, Microsoft Internet Explorer, Netscape Navigator, Safari, Flock, Opera, Google Chrome, etc.

Whenever an Internet connection is established and a web browser is launched, it displays the browser's homepage. The browser's home page can be changed anytime. The term homepage refers to the first web page of a website. Most graphic-based browsers have this common user interface. The common features include:

- Back and forward buttons to go back to the previous web pages and forward respectively.
- A refresh or reload button to reload the current web page.
- A stop button to cancel loading the web page. In some browsers, the stop button is merged with the reload button.
- An address bar to input the Website /Internet address or Uniform Resource Locator (URL) of the desired Website and display it.
- A search bar to input terms into a search engine. In some browsers, the search bar is merged with the address bar.

Basic User Features of a Web Browser Interface



7.1 Why Effective Searching?

There are several reasons for which one must carry out an effective search on the Internet. Some reasons may include:

- There are a huge amount of resources online. For instance, Google claims to have indexed more than 8 billion web pages (Gulli and Signorini, 2005).
- The World Wide Web is not indexed or catalogued, hence we need to formulate our searches carefully.
- We need to make effective use of resources such as computers, bandwidth, time and money. According to Shelly and Vermaat (2011), sharing resources on a network saves time and money.

7.2 Types of Information on the Internet

The World Wide Web has various types of information available online for downloads. These include:

- Publications - grey literature eg. Public sector publications, news articles and magazines
- Web documents - Individuals' and organisations' home pages
- Online journals – electronic journals
- Online books – electronic books
- Reference resources - dictionaries, encyclopedias, etc.
- Images -pictures, diagrams, and drawings
- Videos –documentaries, home or movie videos, music videos, and television programs
- Audio - music, audio recordings and sounds
- Maps - maps of countries or cities, weather maps, and driving directions to a destination
- People or Business directories - addresses and telephone numbers
- Blogs - specific opinions and ideas of others

7.3 Internet Search Tools

Searching for information on the Internet can be an exciting experience if one knows how and which tools to use. There are various resources which can be used to search or retrieve information for academic or research purposes. Most search tools

...slightly in how they function and what information is included in their database. The four most widely used search tools are the Search engines, Metasearch engine, Web Directories/ Portal (Subject Gateways) and Databases:

7.3.1 Search Engines

A search engine is an information retrieval system designed to help find information stored on a computer system (often the World Wide Web). Search engines are made up of programs ("spiders" or "robots") to collect information; to index this information; and create databases of web sites visited which are regularly updated. They have search interface. Some examples are www.google.com, www.bing.com, www.msn.com, www.alltheweb.com, and www.ask.com. Search engines help in locating information for which you do not know the exact web address. They require that you enter a word or phrase, called search text or search query, which describes the item you want to find. Each word in the search text is known as a keyword. Your search query can be broad or more specific. The search engines often respond with thousands or millions of results, whose content varies depending on the type of information you are looking for. Some results have links to Web pages or articles; other results may be images or videos. You may find that many items that appear in the search results may have little or no bearing on the item you are looking for. You can eliminate the unessential items in your search results by carefully choosing search query that limits the search. If you misspell search text, many search engines can identify alternative search text. Some also provide suggested

keywords as you begin typing your search query. The results of the search, called hits are linked to web pages that have your search text in them. Each listed result when clicked, displays the full content of the website or web page.

7.3.2 Metasearch/ Multi-Search Engines

Metasearch engines query several search engines at the same time and combine their results. For example, www.yippy.com generates an ordered list based on comparative ranking. This "metasearch" approach helps raise the best results to the top and push search engine spam to the bottom. Metasearch engines permit users to enter search query once and access several search engines simultaneously. Metasearch engines operate on the premise that, the Web is too large for any one search engine to index it all, and that more comprehensive search results can be obtained by combining the results from several search engines. This also saves the user's time from having to use multiple search engines separately. While some metasearch engines like www.webcrawler.com and www.metacrawler.com query two or three search engines at the same time, others like www.ixquick.com queries over ten different search engines simultaneously. Also, some metasearch engines such as www.dogpile.com do show the search engines where the results are picked from whereas others like www.mamma.com do not do that.

7.3.3 Web Directories/Portals (Subject Gateway or Multi-subject Directory)

Subject Gateway or Multi-subject directory collect and categorize Internet resources (websites and Internet publications of particular interest) into several organized categories or disciplines. Each category contains links to sub-categories that link web pages containing information on broad topics. They do not search the entire Internet. A subject gateway is a useful guide to start your search as it often provides structured overviews of resources. There are commercial subject directories like www.yahoo.com and www.lycos.com that have organized categories such as sports, shopping, weather, etc. On the other hand, Academic subject gateways like ipl2 (www.ipl.org/div/subject), Pinakes(www.hw.ac.uk/libwww/irn/pinakes/pinakes.html) and www virtual library (www.vlib.org.uk) have their organized categories in subject areas such as Agriculture, Business and Economics, Law, Humanities, etc. Academic subject gateways provide high quality information as they are organized by Subject Specialists or Librarians. To use this search tool, you locate a particular subject by clicking links through different levels, moving from the general to the specific. Each time you click a category link, the subject directory displays a list of sub-category links, from which you can select what you need. You continue in this fashion until the search tool displays a list of Web pages on the desired topic.

7.3.4 Databases

A database is simply a collection of structured records. It is structured in such a way to permit orderly retrieval for research, study or administration. A database record contains defined fields. Shelly and Vermaat (2011) defined a database as a collection of data organized in a manner that allows access, retrieval, and use of that data.

Types of Databases

Databases can be categorized as follows:

- Bibliographic Database: It is a database of bibliographic records, an organized digital collection of references to published literature, including journal and newspaper articles, conference proceedings, reports, government and legal publications, patents, books, etc. A bibliographic database may be general in scope or cover a specific academic discipline.
- (eg. MEDLINE, AIDSLINE, HealthSTAR)
- Full-text Databases: These contain the full text of a work and not the abstract only, which are presented right on the screen.
- Numeric Databases: These provide mostly numeric data such as statistics, financial data, census information, and economic indicators (eg. www.technet.microsoft.com)
- Image Databases: These are databases which contain graphics (eg. www.flickr.com, Google images, etc)
- Audio Databases: These are databases which only contain collections of audio data (eg. www.spotify.com)

- Video Databases: These are databases which only contain collections of video data (eg. www.youtube.com)
- Mixed Databases: These are databases which contain a mixture of information in text, graphic and audio/video (eg. www.google.com)

Academic Databases

Academic database is a universal index of periodical literature covering basic research from all fields of knowledge. They contain records with defined fields such as title, author, date, URL, etc. Academic Journals Database contains complete bibliographic citations, precise indexing, and informative abstracts for papers from a wide range of periodicals. Some academic databases are free to use while some may require subscription to access its content. Some examples are Highwire, JSTOR, SAGE, PubMed, Hinari, Agora, OARE, Science Direct, Emerald, and Ebsco. Academic databases provide high quality information as their contents have been evaluated or peer reviewed, thereby providing scholarly journal articles.

7.4 Searching the World Wide Web

Searching the World Wide Web can be an overwhelming task. One must however know the 'basic steps in order to search effectively. To effectively retrieve information from the Internet, one can employ the following steps discussed below:

1. Define your information need
2. Choose your search terms
3. Decide which sources/search tool to use

4. Find out how the search tool functions
5. Run your search
6. Review and refine your search

Note: Steps 1-3 can be done offline or away from a computer connected to the Internet.

1. Define Your Information Need

- What sort of information are you looking for?
  – Is it specific information? eg. a fact or date;
  - Reference source, e.g. data book, encyclopaedia, dictionary, the Web or even a textbook could be the best source.
  - General information, eg. research topics which may require more thought, including how much information is needed and at what depth
- Who is going to use the information?
  – Researcher? First year student?
  - This might affect the sort of information you require

2. Choose Your Search Terms

- What unique words do you think will appear in the site/article you want?
- Are there any key phrases?
- Are there any synonyms, alternative spellings, plurals or capitals that you need to consider
- What broader topic is it a part of or related to?

Example: I want to find information about the health implications of water pollution

- Keywords–'water' 'pollution' 'health'
- Broader topics–'environmental degradation' or 'agricultural management' or 'health'
- Synonyms:
  – Water: rivers, lakes, sea, coastal, 'domestic water', etc.

3. Decide the Sources/ Search Tool to Use

- What sources or search tool are appropriate for your information need?
  – Search engines, Subject gateways, academic databases, etc.
  – Reference resources, e.g., encyclopaedias, dictionaries
  – Individuals' and organisations' home pages
  – Newspapers, etc.

4. Find Out How the Search Tool Functions

All electronic search tools function slightly differently and may use some of the following functions:

- Phrase searching: allows you to search for an exact phrase, and it is used for performing simple searches.
- Boolean operators: using AND, OR, NOT in performing advance searches
- Truncation/wildcards: allows you to retrieve documents containing variations on a search term. To do this, type the first few letters (stem) of the keyword followed by an asterisk (*). For example, creat* would retrieve entries for all

documents containing the words: creature, creation, create, creating, creator, etc.

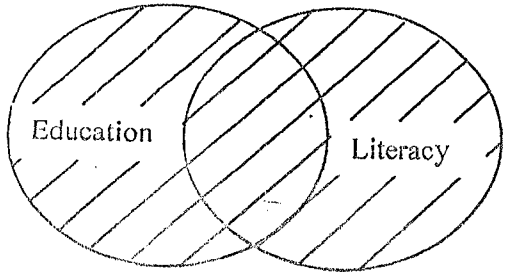
- Case sensitivity: recognition of upper or lower cases in search terms
- Fields: searches in fields such as the title, URL or links
- Stop words: searches may ignore terms such as: and, if, an, the

Note: Read the 'Help' page to find out how that particular search tool functions (i.e, uses Boolean operators, wildcards, etc.) by going to the homepage, enter a simple search query, and then click on "Help" usually at the bottom of the page.

Boolean searching is the conventional basis for searching most computerized systems, including bibliographic databases. Most search engines allow you to combine terms with words referred to as Boolean operators such as "and," "or," or "not." Knowing how to use these terms is very important for a successful search. Most search engines will allow you to apply the Boolean operators in an "advanced search" option.

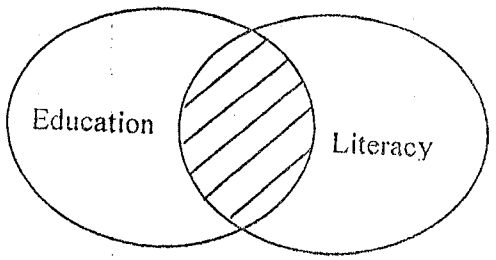
- Uses commands (operators) such as AND, OR, NOT
- Different search tools may use different symbols
  AND (+)
  NOT (–)
- Different search tools may use OR or AND as a default setting
- Sometimes Boolean operators must be entered in capital letters

OR is an operator that allows either word to be present in each record in the results. You use OR operator to expand your search. For example, to enter a search query 'Education OR Literacy' means you want the search tool to provide results on either of the two keywords. Hence, the results provided will cover information from websites which contain the word education only or literacy only or both education and literacy. The venn diagram below illustrates the function of the OR operator by shading all sections of the circles.



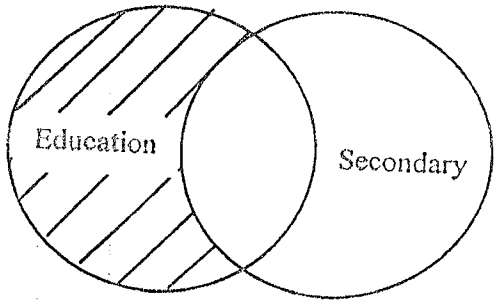
Query: I would like information on education or literacy

AND is an operator that requires both words to be present in each record in the results. You use AND operator to narrow your search. For example, to enter a search query like 'Education AND Literacy' means you want the search tool to provide results on both keywords. Therefore, the results displayed will cover information from websites or articles which contain both education and literacy, ignoring any information on only education or only literacy. The venn diagram below illustrates the function of the AND operator by shading the intersection of the circles.



Query: I am interested in the relationship between education and literacy

NOT is a connector that requires the first word to be present in each record in the results, however, it excludes the second word. For example, to enter a search query on 'Education NOT Secondary' means you want the search tool to display results on only the first word, which is education and ignore all sites or articles which have either the word secondary or both education and secondary in them. The shaded part in the venn diagram below demonstrates how the NOT operator works.



Query: I want to see information about education, but I want to avoid seeing anything about secondary.

of these words" implies the operator AND, whereas "Any of th words" implies the operator OR.

5. Running the Search

- Take the terms/keywords you have decided on and enter the in the search bar of the search tool.
- Run the search by clicking on the search button.
- Search results will be displayed with hyperlinks after th search. Hyperlinks are connections to other documents or we pages that contain related information. They appear a underlined or coloured words and phrases.

6. Review and Revise Your Search

- The information or results displayed must be evaluated fo quality and authenticity
- The search scope (query) and approach can be revised
- Try new sources of information

7.6 Evaluating Information from the Internet

The Internet is a large electronic network which has n centralized monitor or control; and is however not regulated fo reliability, authority, or accuracy. When research is conducted o the Internet for academic or research purposes, it is important t evaluate the credibility of the websites you visit. Critically, th author's credentials, educational background, expertise on th topic, the author's association with a credible institution, etc. ar very relevant to check. It is therefore prudent that the criteri

applied in evaluating print sources of information should be applied in evaluating information retrieved from the Internet. Some aspects to consider are:

Reliability

Information taken from the Internet should be examined by checking the source if it could be relied upon. This can be done by examining the Website's URL if the source is from an educational institution (.edu), a government agency (.gov), a non-profit organisation (.org) or a commercial enterprise (.com), etc. If the source is from an educational institution, is there a disclaimer saying that the institution is not responsible for the contents? You can also check listed sources, contact information, available collaboration, claims supported, and documentation supplied on the site. The goal of this is to be sure that the source provides convincing evidence for the claims made.

URLs provide a standard format for the transmission and reception of a wide variety of information types. Here is how they are constructed:

Transferprotocol://servername.domain/directory/subdirector y/filename.filetype

Example:http://www.nps.gov/grsm/planyourvisit/wildlifeviewing. htm.

...major criterion used in evaluating information. ...hen we look for information with some type of critical value, we want to know the basis of the authority with which the author speaks. Here are some possible factors to consider:

- In your field of study, is the author well-known and well-recognized.
- If you find an author you do not recognize, check his credentials, expertise on the topic and educational background by reading the biographical information including the author's position, institutional affiliation and address (telephone number and email in order to request further information on his or her work and professional background).

Currency/Timeliness

This refers to the timeliness of the information. In printed documents, the date of publication is the first indicator of currency. On the Internet, one can check the document date which includes the publication date or the "last updated" date or date of copyright. If no date is given in an electronic document, you can view the directory in which it resides and read the latest modification date, or you can check the document's bibliography or references dates provided.

Accuracy/Completeness

Information for research or academic purpose must be complete and objective with relevant supported facts, and not subjective. It

is possible to have postings on the Internet which are just excerpts or summaries of another work in a printed source. It is also possible that information may be solely reflective of one's opinion. Every writer wants to prove his point, and will use the data and information that assists him in doing so. When evaluating information found on the Internet, it is important to examine who is providing the "information" you are viewing, and what might be their point of views or biases. The following could be checked to evaluate accuracy:

- The methodology outlined in the document should be appropriate to the topic and allows the study to be duplicated for purposes of verification.
- The information should list other sources relied on in the bibliography or includes links to the documents themselves.
- The information should name individuals and/or sources that provided non- published data used in the preparation of the study.
- The background information that was used can be verified for accuracy

Relevancy

It is important to remember that sources of information comprising the Library's print and electronic collections have already been evaluated for inclusion among the Library's resources. However, this does not necessarily mean that these sources are relevant to your research. Relevance refers to how well the information meets your research needs. To determine

information's relevance, consider answering the question "how closely does this information relate to my research question?" When searching for information, it is important to remember that all of the information that you use should closely address your topic. If the information barely talks about your topic, then it is probably not something that you should use.

Credibility

With respect to print sources, the publishing body or publisher also helps to evaluate any kind of document that one may be reading. This generally means that the author's manuscript has undergone screening in order to verify that it meets the standards or aims of the organisation that serves as publisher. This may include peer review. Similarly, on the Internet the trustworthy source or credibility of the institution or organisation hosting the information provides an authoritative source that supplies some good evidence that allows one to trust the information. However, one must ask the following questions to assess the role and authority of the hosting institution or server (computer) where the document is stored:

- Does the website clearly identify the organisation responsible for publishing the information found on it?
- Is there a link at the top or bottom of the page linking back to the information about the website's publisher or sponsor?
  Note: You can often find such information from an "About Us" or "Frequently Asked Questions" link.

7.7 Information and Web 2.0 Technology

The term "Web 2.0" (2004–present) is commonly associated with web applications that facilitate interactive information sharing, interoperability, user-centered design and collaboration on the World Wide Web. Examples of Web 2.0 include web-based communities, hosted services, web applications, social-networking sites, video-sharing sites, wikis and blogs. A Web 2.0 site allows its users to interact with other users or to change website content, in contrast to non-interactive websites where users are limited to the passive viewing of information that is provided to them. The term is closely associated with Tim O'Reilly because of the O'Reilly Media Web 2.0 conference in 2004. Although the term suggests a new version of the World Wide Web, it does not refer to an update to any technical specifications, but rather to cumulative changes in the ways software developers and end-users use the Web. Whether Web 2.0 is qualitatively different from prior web technologies has been challenged by World Wide Web inventor Tim Berners-Lee, who called the term a "piece of jargon" — precisely because he intended the Web to embody these values in the first place.

Seeking and retrieving information on-line has become more interactive by the introduction of web 2.0 services and technologies. The individual is not always the dormant receiver of the information on the web, but could make their views public by these tools. It offers opportunity for information and knowledge exchange, and collaboration among scholars

Web 2.0 services and technologies include content generation and collaboration:

Wikis, Blogs, Google Docs

Content aggregation

RSS, Google Reader

Sharing of documents online

Scribd, SlideShare

Office information management

Backpack, Stikkit

WIKI

Wiki is the shorten form of the Hawaiian word for quick. This is a page or collection of Web pages designed to enable anyone who accesses it to contribute or modify content, using a simplified markup language.

Types of Wikis

- Public – anyone can edit
- Private – access can be restricted to small or large groups
- Hybrids – public viewing but editing is restricted on some or all sections

Uses of Wikis

- Collaborative work - planning, brainstorming, writing articles & documents, etc
- Internal documentation – manual, operational guidelines, technical documents, etc
- Dashboard - showing deadlines, things to focus on, etc

- Fact and FAQ lists
- Project Management
- Corporate glossaries and dictionaries

Advantages of Wikis

- Anyone can edit or add text, and content is subject to peer review
- Allows community ownership of information - (i.e. no longer belongs to one person).
- Employees may come and go, but you keep the knowledge in the firm
- Allows community ownership of information - (i.e. no longer belongs to one person).
- Increases the collective intelligence

Disadvantages of Wikis

- Overcoming cultural hurdles of hierarchy, control, and a sense of lack of accountability
- Issues of legal liability, privacy, reputation, and security must be considered
- Concerns about ownership of content
- Concerns with accuracy

While the overall trend is toward improvement, it is important to use Wikipedia carefully if it is intended to be used as a research source, since individual articles will, by their nature, vary in quality and maturity. Guidelines and information pages are

available to help users and researchers do this effectively, as is an article that summarizes third-party studies and assessments of the reliability of Wikipedia

Examples of Wikis

- Wikipedia

http://en.wikipedia.org/wiki/Wikipedia

- Adobe Wiki

http://adobe.wikia.com/wiki/Adobe_Wiki

BLOG

Blog is a personal or corporate website in the form of an online journal, with new entries appearing in sequence as they are written, especially as dealing with reflections or opinion, and typically incorporating links to other articles. The term "blog" is derived from "Web log."

Blogger - A person who writes a blog

Blogosphere - The online community of blogs and the people who write them

Blogroll - A list of links to blogs recommended by the blogger and typically found in the blog's sidebar

Features of Blogs

- A main content area with articles listed chronologically, newest on top
- An archive of older articles
- A way for people to leave comments about the articles.
- A list of links to other related sites

- One or more "feeds" like Rich Site Summary(RSS), Atom or Resources Description Framework (RDF) files

Uses of Blogs

- Personal journal
- Marketing
- Knowledge sharing
- What's new news

Advantages of Blogs

- An excellent way of sharing knowledge within the organisation;
- A positive way of getting feedback;
- Can build profiles of the writers;
- Showcases the organisation's talent and expertise.

Disadvantages of Blogs

- Blogs are hard to maintain
- Writers may present different views from those of the organisation

GOOGLE DOCS

What is Google Docs?

- Web-based application that allows users to prepare documents, spreadsheets, and presentations through their browser

- Online workspace
- Facilitates collaboration:
  - Creating documents
  - Viewing documents
  - Editing documents

Features of Google Docs

- Upload documents from and save to your desktop
- Edit anytime, from anywhere
- Pick who can access your documents
- Share changes in real time
- Files are stored securely online
- It is free!

Uses of Google Docs

- Co-authoring articles, reports, manuals, etc
- Personal document storage space
- Develop/brainstorm on ideas
- Collaborative projects

Advantages of Google Docs

- Work on documents from any computer with Internet connection
- Collaboration—multiple editors, simultaneously
- Changes/edits are seen by everyone immediately
- Automatically saves
- Can "revert" to previous versions

Disadvantages of Google Docs

- Need to have a computer with an Internet connection in order to access your documents
- Slow speed of Internet connection can be a challenge

RICH SITE SUMMARY (RSS) OR REALLY SIMPLE SYNDICATION

- Resource Description Framework (RDF) Site Summary

What is RSS?

A format for delivering regularly changing web content
Source: http://www.whatisrss.com/
A standardized dialect of XML that allows content from one site to be republished (syndicated) by another
Source: http://www.rss-syndication.com/

RSS is similar to traditional newspaper syndication—where one company creates a news article and many newspapers carry it—but updated for the Web

What are RSS Feeds?

- A feed contains a list of items or entries, each of which is identified by a link. Each item can have any amount of other metadata associated with it as well
- RSS feeds are plain text (XML) files designed for computer-to-computer communication

Uses of RSS

- To keep current with happenings and trends in ones fields of interest
- To deliver information and news to stakeholders conveniently
- To keep websites current and fresh
- To provide listings of past or upcoming events, deadlines or holidays
- To provide headlines, notices and any list of announcements
- To promote websites

Advantages of RSS

- Saves time by not needing to visit each web site individually
- Improves the site's visibility
- Increases site's popularity with search engines
- Attracts traffic to the site

SCRIBD

What is Scribd?

A type of a social networking website that allows users to publish and share all kinds of documents and images on the web

Some Facts about Scribd

- Uses iPaper to view documents
- Allows users to comment on the documents
- Restriction on access to documents (Private)
- Users can flag a document (report if it infringes the set rules)

- An account can be terminated for more than two reported instances of copyright infringement, among other things

Advantages of Scribd

- You can access your documents from any computer with Internet connection
- Facilitates sharing of documents
- Comments on documents can be very valuable

List of Web 2.0 Tools and Services Websites

1. Tools for Web Content Generation and Collaboration
   - PBWiki (http://pbwiki.com)
   - Wikispaces (http://www.wikispaces.com/)
   - WordPress (http://wordpress.org)
   - Blogger (https://www.blogger.com/start)
   - Wetpaint (http://www.wetpaint.com/)
   - Google Docs (http://docs.google.com)

2. Tools for Office Information Management (i.e. Notes, Calendars, Schedule of Meetings, Things to do Lists)
   - Remember the Milk (http://www.rememberthemilk.com/)
   - Stikkit (http://www.stikkit.com/)
   - Backpack (http://www.backpackit.com/)

3. Tools for Resource Sharing (multimedia – photos, videos and podcasting)
   - YouTube (http://www.youtube.com)

- Flickr (http://www.flickr.com)
- Podcast Alley (http://www.podcastalley.com/)

4. Tools for Resource Sharing (Documents, Spreadsheets, Presentations)
   - Scribd (http://www.scribd.com/)
   - SlideShare (http://www.slideshare.net)

5. Tools for Social Networking
   - Facebook (http://www.facebook.com/)
   - Linkedin (http://www.linkedin.com/)

6. Tools for Tagging and Social Book Marking
   - del.icio.us (http://delicious.com/)
   - Yahoo MyWeb (http://myweb.yahoo.com/)
7. Tools for Aggregating Content
   - AgriFeeds (www.agrifeeds.org/)
   - Google Reader (http://www.google.com/reader)
   - RSS

7.8 Massive Open Online Course

A massive open online course (MOOC) is an online course aimed at unlimited participation and open access via the web. In addition to traditional course materials such as filmed lectures, readings, and problem sets, many MOOCs provide interactive user forums to support community interactions between students, professors,

and teaching assistants (TAs). MOOCs are open to anyone in the world. They are free to sign up for and the student does not need to be enrolled at the institution offering the MOOC. Courses are delivered online and students can typically work at their own pace.

MOOC has changed the face of traditional education with the introduction of online, or e-learning and distance education, with increasing online presence, open learning opportunities, and the development of MOOCs. Student and instructor participation, motivation, instructional method, and delivery are all important aspects necessary to create a MOOC environment conducive to learning. MOOCs build on the engagement of learners who self-organize their participation according to learning goals, prior knowledge and skills, and common interests.

Benefits of MOOC

1. MOOCs allow participants to pursue a particular interest or to continue their professional development
2. MOOCs offer educational opportunities exist for underprivileged populations as a way to encourage lifelong learning.
3. They provide students the opportunity to expand their knowledge and pursue their interests without dedicating fixed periods of time to fit a college or university schedule.

4. Employers can utilize MOOCs to keep employees abreast of the competitive labor market throughout their lifetime and in a way that is cost-effective.
5. They offer opportunity for great collaboration between the worlds of higher education and corporate learning, though they may be miles apart from one another.

Challenges of MOOC

1. It relies on user-generated content which has the potential to create a chaotic learning environment
2. Digital literacy is necessary to make use of the online materials.
3. The time and effort required from participants may exceed what students are willing to commit to a free online course.
4. Once the course is released, content will be reshaped and reinterpreted by the massive student body, making the course trajectory difficult for instructors to control
5. Participants must self-regulate and set their own goals

Some MOOC providers:

Provider	Type	Website Address and Participating Institution	Headquarters	Founded
Stanford Online	Non-profit	Stanford University http://online.stanford.edu/	USA	2006
Khan Academy	Non-profit	https://www.khanacademy.org/	USA	2006
WizIQ	Commercial	IIT Delhi (http://www.iitd.ac.in), Des Moines Area Community College www.dmacc.edu	India/USA	2007
Canvas Network	Commercial	Santa Clara University(www.scu.edu), University of Utah www.utah.edu	USA	2008
Peer to Peer University	Non-profit	https://p2pu.org/en/	USA	2009

Academic Earth	Non-profit	UC Berkeley(www.berkeley.edu) UCLA(www.ucla.edu) Oxford University(www.ox.ac.uk)	USA	2009
Udemy	Commercial	University of Chicago Law School www.law.uchicago.edu George Washington University www.gwu.edu	USA	2010
Udacity	Commercial	Georgia Institute of Technology www.gatech.edu San Jose State University www.sjsu.edu	USA	2012
Eliademy	Commercial	Aalto University Executive Education www.aaltoee.com	Finland	2012
openHPI	N/A	Hasso Plattner Institute https://open.hpi.de/	Germany	2012

Source: Wikipedia

References

Beekman, G. & Quinn, M. J. (2008). *Tomorrow's technology and you* (8th ed.). New Jersey: Prentice Hall.

Database search tips: Boolean operators. Retrieved from http://libguides.mit.edu/content.php?pid=36863&sid=271372 on 20th July, 2013.

Evaluating information found on the Internet. Retrieved from http://guides.library.jhu.edu/content.php?pid=198142&sid=1665954 on 26th July, 2013.

Evaluating sources. Retrieved from http://knowledgecenter.unr.edu/help/manage/evalsources.aspx on 26th July, 2013.

Evaluating web information. Retrieved from http://www.lib.unca.edu/library/lr/evalweb.html on 26th July, 2013.

Glister, P. (1997). *The web navigator*. New York: Wiley Computer Publishing.

Gulli, A. & Signorini, A. (2005). Building an open source meta search engine. WWW2005 conference, May 10-14, Chiba, Japan.

International Network for the Availability of Scientific Publications (2000). Workshop materials on Internet use. Oxford: INASP.

International Network for the Availability of Scientific Publications (2002). Workshop electronic journals and electronic resources library management. Oxford: INASP.

Kellogg, S. (2013). Online learning: How to make a MOOC. *Nature*, 499(7458), 369-371.

Lever-Duffy, J. & McDonald, J. B. (2008). *Teaching and learning with technology* (3rd ed.). New York: Pearson Education Inc.

Massive Open Online Course. Retrieved from https://en.wikipedia.org/wiki/Massive_open_online_course on 20th March 2015.

Shelly, G. B. & Vermaat, M. E. (2011). *Discovering computers 2011: Living in a digital world, complete.* Boston: Cengage Learning.

Wu, K. (2013). Academic Libraries in the age of MOOCs. *Reference Services Review*, 41(3), 576-587.

AUTHOR INDEX

Afful-Arthur, P. 15, 35
Aina, L.O. 15, 34, 36, 41, 57, 60
Alan, D. 60
Alemna, A.A. 15, 21, 22, 86
Amanquah, S.N. 79
Asamoah-Hassan, H.R 99
Baiden, W.B. 61, 80
Baumbach, D.T. 4, 14
Beekman, G. 153
Bolner, M.S. 79
Berners-Lee, T. 139
Biney, W.K. 61, 80,101, 120
Boohene, K.A. 101, 120
Bopp, R.E. 45, 60
Borgman, C.L. 34, 60
Donahue, G.A. 118
Donald, B.M. 99
Farrow, J. 79
Filson, C.K. 15, 35
Gates, J.K. 34
Glister, P 153
Gulli, A. 122, 153
Joudrey, D.N. 79
Katz, W.A. 52, 60
Kellog, S. 154
Kofi, C.Y. 1, 61, 80
Kwafoa, P.N.Y. 101, 120
Laryea, K.A. 15, 35
Lea, P.W. 60
Lever-Duffy, J. 154

Loertscher, D.V. 4, 14
Long, C. 118
Mann, M. 73, 79
Mayank, T. 34
McDonald, J.B. 154
Montgomery, P. 4, 14
Morris, B.J. 20, 34
Nunekpeku, P. 101, 120
Oja, D 118
Opare- Adzobu, J.A. 15, 35
O'Reilly, T. 118
Osei-Agyemang, D. 61, 80
Parsons, J.J. 118
Poirer, G.A. 79
Quinn, M.J. 153
Reitz, J.M. 40, 43, 60
Rowley, J.F. 79
Sawyer, S.C. 119
Sayers, W.C. 73, 79
Shelly, G. B. 103,104,118,122,127
Signorini, A. 122, 153
Smith, A.N. 99
Smith, L. C. 60
Taylor, A.G. 79
Taylor, W. 79
Thomas, N.P. 3, 4, 10, 14
Vermaat, M.E. 103, 104, 123,127, 118
William, H.K.R. 79
Williams, B.K. 119
Wu, K. 154

SUBJECT INDEX

A
Abstracting Services 25
Abstracts 56
Abstracts Types 57
Abstracts Uses 59
Aburi Botanical Gardens 22
Academic Databases 128
Academic Libraries 18
Academic Library- Collections 18
Academic Library - Electronic Support 21
Academic Library – Lending Service 20
Academic Library – Reference Service 20
Academic Library – Reprographic Service 21
Accuracy 11, 136
Academic Library - User Education 22
Access Points 71
Acquisitions 15
African Studies Database 58
Akosombo International School 17
American Library Association 4, 14
Anglo-American Cataloguing Rules (AACR) 67, 65
APA Reference List 86-94
ARPANET 103,104
Association of College & Research Libraries 5, 14
Atlas 50
Atlas Types 50
Atlas Uses 51

Author Bibliograhies 45
Authority 11, 136

B
Bandwidth 109
Bias and Accuracy 11
Bibliographies 44
Bibliography Types 45
Bibliography Uses 46
Biographies 47
Biography Types 47
Biography Uses 48
Blog 142
Blog - Advantages 143
Blog -Disadvantages 143
Blog –Features 142
Blog –Uses 143
Boolean Searching 131
Book Catalogue 69
Book Trade Bibliography 46
British National Bibliography 46
Broadband Connections 105
Browsers 120

C
Cable Modern Connection 108
Call Mark 71
Card Catalogue 69

Cataloguing 62
Cataloguing Description 65
Cataloguing Record 64
CD – ROM 17,19, 21, 32,39,46,55

Chartered Institute of Library & Information Professionals 14
Citation 80
Citation – Styles 83
Citation – Types 84
Classification 73
Classification System 74
Collections 17,18,23,26
Communication 112
Computer Access Catalogue 70
Computer Networks 101
Computer Virus 117
Concept Map 13
Consortium of Academic & Research Libraries in Ghana (CARLIGH) 37
Context 12
Copyright 13, 80
Copyright Law 80
Council for Scientific & Industrial Research (CSIR) 23, 58
Coverage 11
Credibility 11
Critical Thinking 1, 9, 10

Currency and Timeliness 136
Cut & Paste 13

D

Databases 127
Data protection 13
Dewey Decimal Classification Scheme 74
Desktop Computer 110
Dial-up Connection 105
Dictionaries 38
Dictionary Types 38
Dictionaries Uses 40
Digital Duplicating Device 21
Digital library 31
Digital Subscriber Line 106
Directional Search Tools 35
Directories 43
Directories Types 43
Directories Users 43
Dissertation Abstracts 58
Document Delivery 21

E

E-Books 19, 33
E-Books Reader 111
E-Commerce 114
Education Research 113
Electronic Library 31
Electronic Mail (Email) 8, 20, 112

Electronic Support 21
Encyclopaedias 40
Encyclopaedia Types 41
Encyclopaedia Uses 42
Entertainment 114
Ethical Issues 80
Evaluating Information Sources 10
Exotic Technologies 8

F

Fair Use 13, 21,
Forms of Libraries 30
Fraud 116
Freedom of Information 13

G

Gazetteers 52
Gazetteers- Uses 52
General Index 55
George Padmore Research Library on African Affairs 29
Geographical Sources 49
Ghana International School 17
Ghana Library Authority 26, 27
Ghana National Bibliography 46
Ghana Science Abstracts 58
Ghana Tourism Authority 53
Google Documents 143

Google Documents Advantages 144
Google Documents Disadvantages 145
Google Documents Features 144
Google Documents Uses 144

H
Harvard Citation Styles 85
Higher Education 9
Hybrid Collections 33
Hybrid Library 33

I
Identity Theft 116
Indexes 55
Index Types 55
Index Uses 59
Indexing Services 25
Indicative Abstract 57
Information 62
Information Age 2, 8
Information Dissemination 113
Information Evaluation 134
Information Evaluation Accuracy 136
Information Evaluation Authority 136
Information Evaluation Credibility 138
Information Evaluation Currency 136
Information Evaluation Relevancy 137
Information Evaluation Reliability 135

Information Gap 2
Information Literacy 3, 7
Information Overload 2
Information Sources 10, 35
Information Super Highway 2
Information Technology 39
Informative Abstract 57
Integrated Library System (ILS) 75
Intellectual Content 17, 63
Intellectual Property 13
Inter-Library Lending & Document Delivery 20
International Standard Bibliographic Description 65
International Standard Book Number 29, 72
International Standard Serial Number 29, 53
Internet 101, 103
Internet Access 104
Internet Broadband 105
Internet Devices 110
Internet Dial-up connection 105
Internet History 103
Information types 123
Internet Misuse 115
Internet Protocols 103
Internet Search Tools 123
Internet Uses 111
Ipad 111

Johannes Zimmermann Library 23
Journal Index 56

K
Knowledge
Koha 62
76

L
Laptop Computers
Legal Citation 110
Legal Deposit 86
Libraries 28
Library Types 15
Library Catalogue 16
Library Catalogue - Forms 68
Library Cooperation 69
Library of Congress 20
Library of Congress Classification Scheme 29,74
Library of Congress Subject Headings 74
Life Long Learning 63
Local Area Network 1,3,
102

M
Machine Readable Catalogue (MARC) 63
Main Entry 65
Maps 49
Massive Open Online Course (MOOC) 148

Massive Open Online Course – Benefits 149
Massive Open Online Course – Challenges 150
Metadata 67
Metadata Types 68
Metasearch Engines 125
Metropolitan Area Network (MAN) 102
Microform 19,30,31,54
Microform Catalogue 69
Ministry of Food & Agriculture 23
MLA Reference List 89
Mobile Broadband 109
Multi-Subject Directory 126
Multi-Search Engines 125

N
National Bibliography 45
National Commission on Excellence
in Education 3
National Library 27
National Library Collections 28
National Library Functions 28
National Library Services 28
Netbook Computer 110
Newspaper Index 56
Non-print Resources 17, 28, 30
Notebook Computer 110

O
Online Databases 21, 89
Online Public Access Catalogue (OPAC) 33, 70
Online Services 114
Organisation of Knowledge 61

P
Periodicals 53
Phishing 117
Plagiarism 80, 95
Pornography 115
Print Resources 17
Problem-Solving 1, 9
Public Libraries 25
Public Libraries Collections 26
Public Library Services 26

Q
Quality 12

R
Radio 8, 49, 106
Reference Materials 36
Reference Materials Characteristics 36
Reference Materials Evaluation 37
Reference Service 18, 20
Responsible Use of Information 13
Resource Description Access (RDA) 67
Rich Site Summary (RSS) 145

S
Satellite Modem Connection 107
Scanner 21, 31
School Libraries 16
School Library Collections 17
School Library Services 18
Scope and Relevance 11
Scribd 146
Search Engines 124
Searching Tools 120
Sears List of Subject Headings 63
Selective Dissemination of Information 24
Smart Phone 111
Spamming 116
Special Libraries 22
Special Libraries Collections 23
Special Libraries Services 24
Subject Abstract 58
Subject Bibliographies 45
Subject Gateways 126
Subject Index 55
Supreme Court of Ghana Library 24

T
Tablet PC 110
Television 8, 49, 106
Thematic Maps 50
Topographical Maps 50

Traditional Library 30
Travel Guide 52
Types of Libraries 16

U
University of Cape Coast 19, 31, 56
University of Education, Winneba 31
University of Ghana 31
User Education 1, 22

W
Web-Based Resources 123
Web Directories 126
Web Portal 126
Web 2.0 Technology 139
Wide Area Network 102
Wiki 140
Wiki Advantages 141
Wiki Disadvantages 141
Wiki Types 140
Wiki Uses 140
Wireless 108
Wireless Fidelity (Wi-Fi) 108
World Wide Web 104
World Wide Web Searching 128